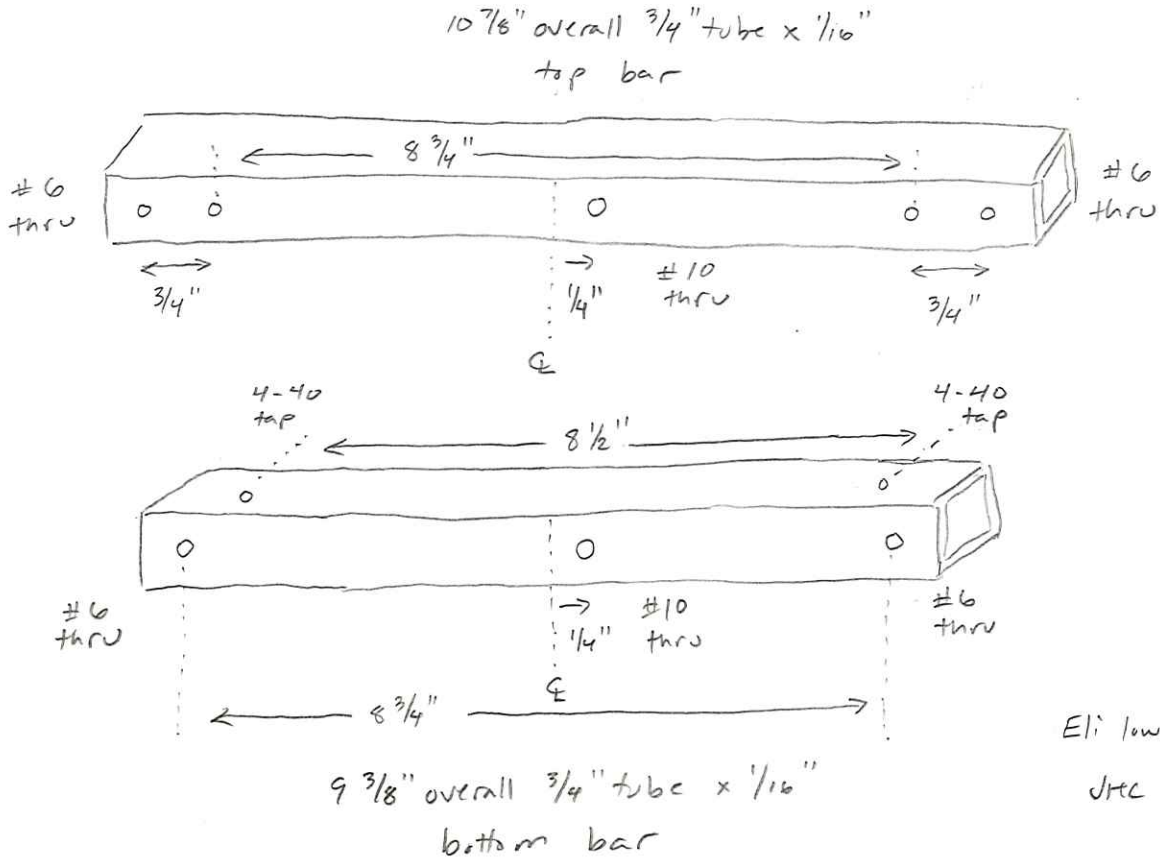
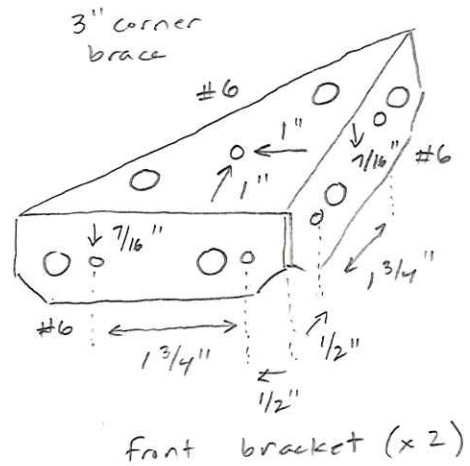
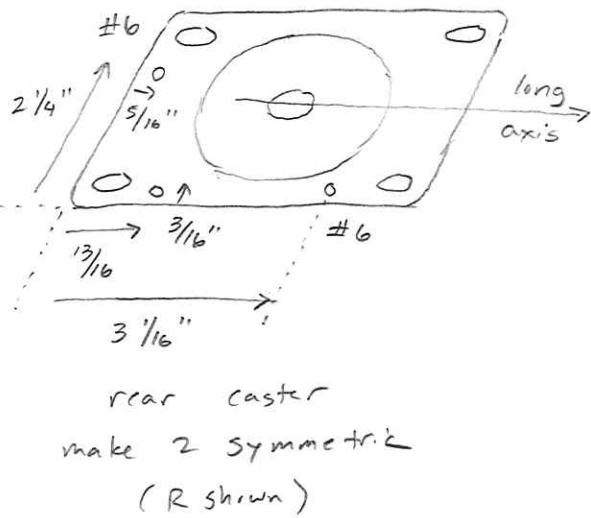
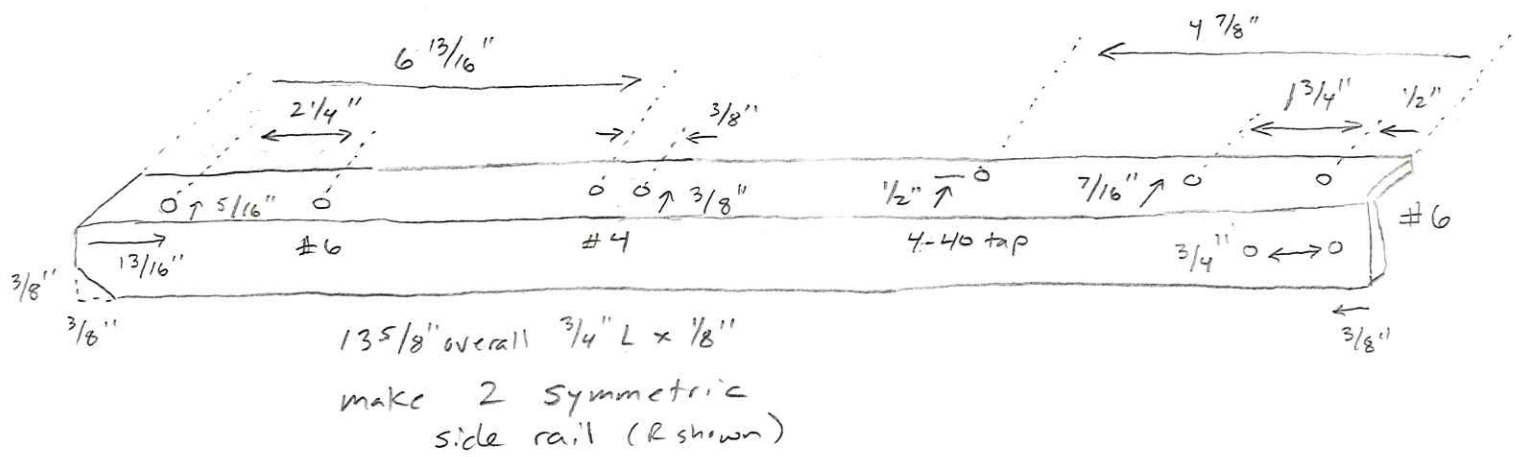
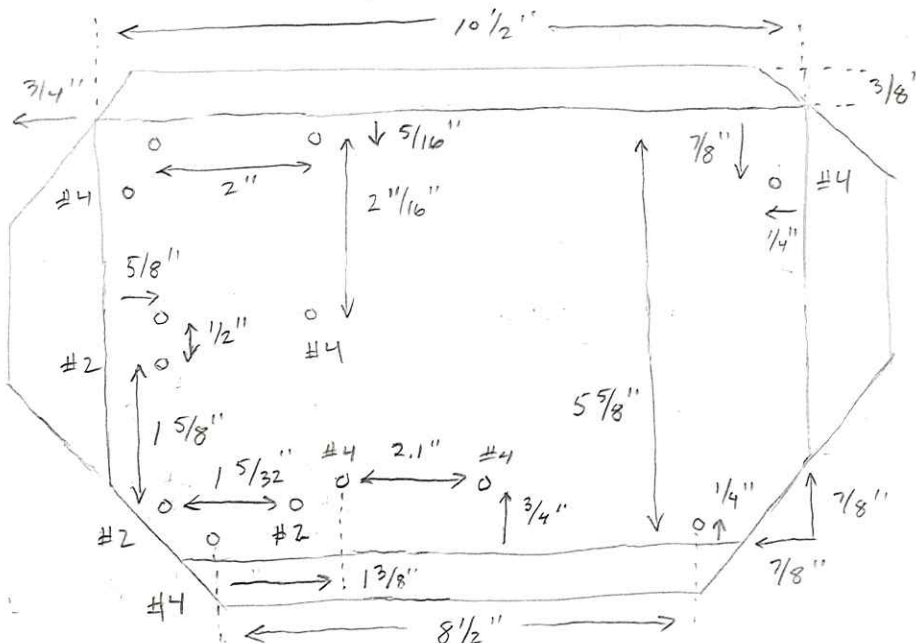
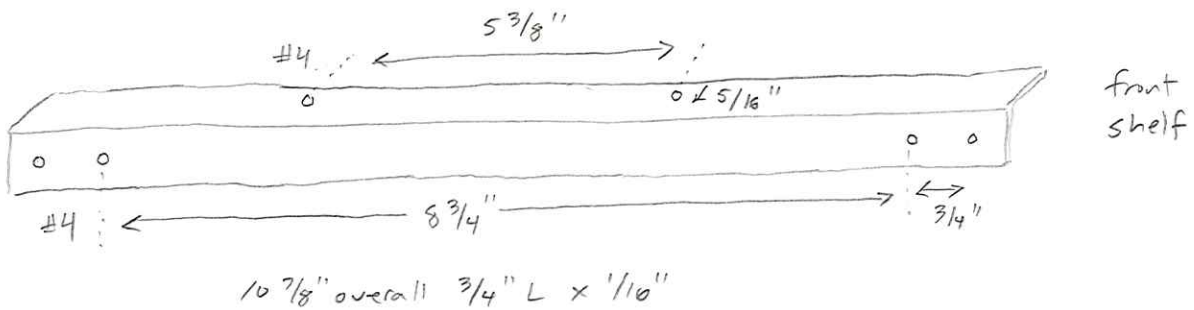
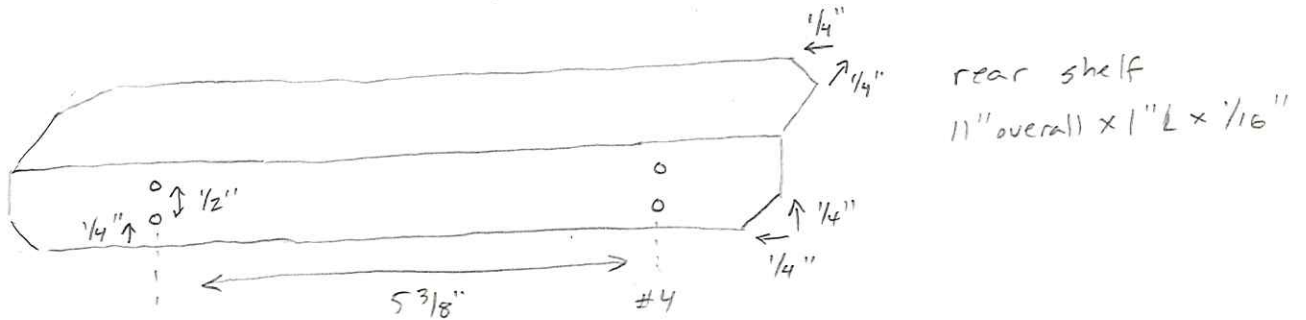
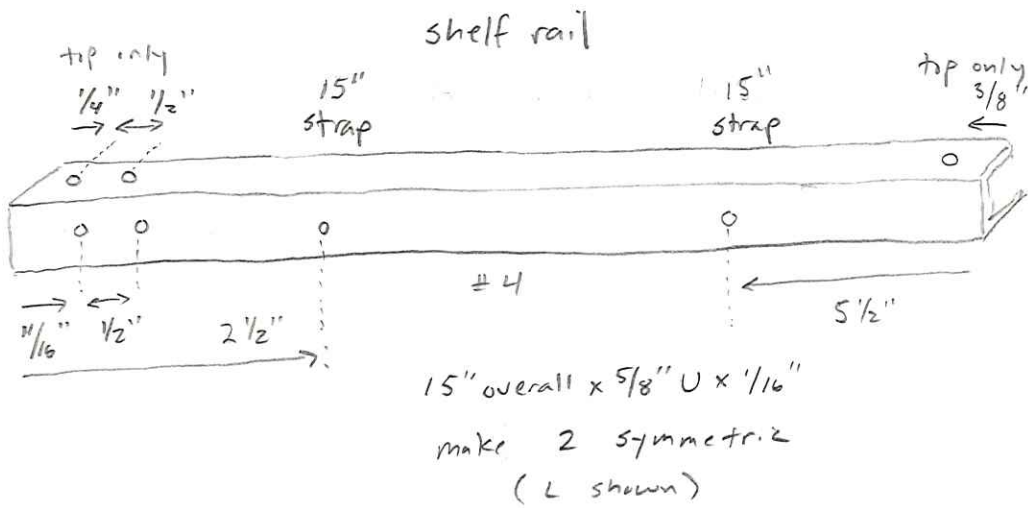


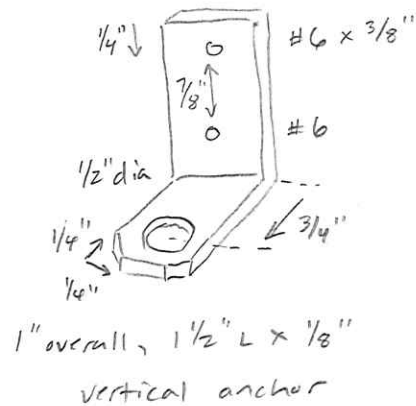
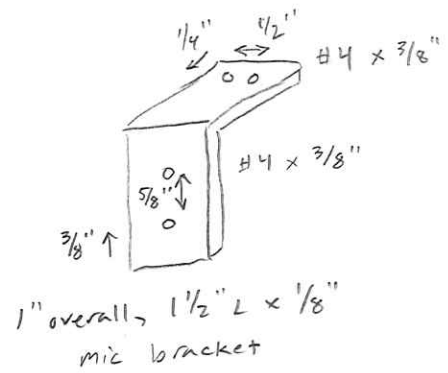
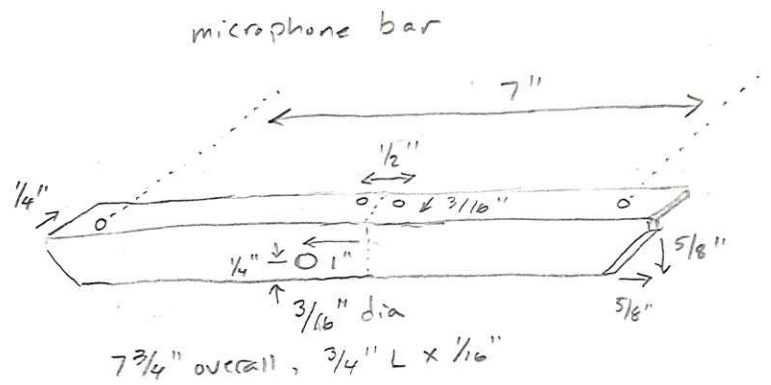
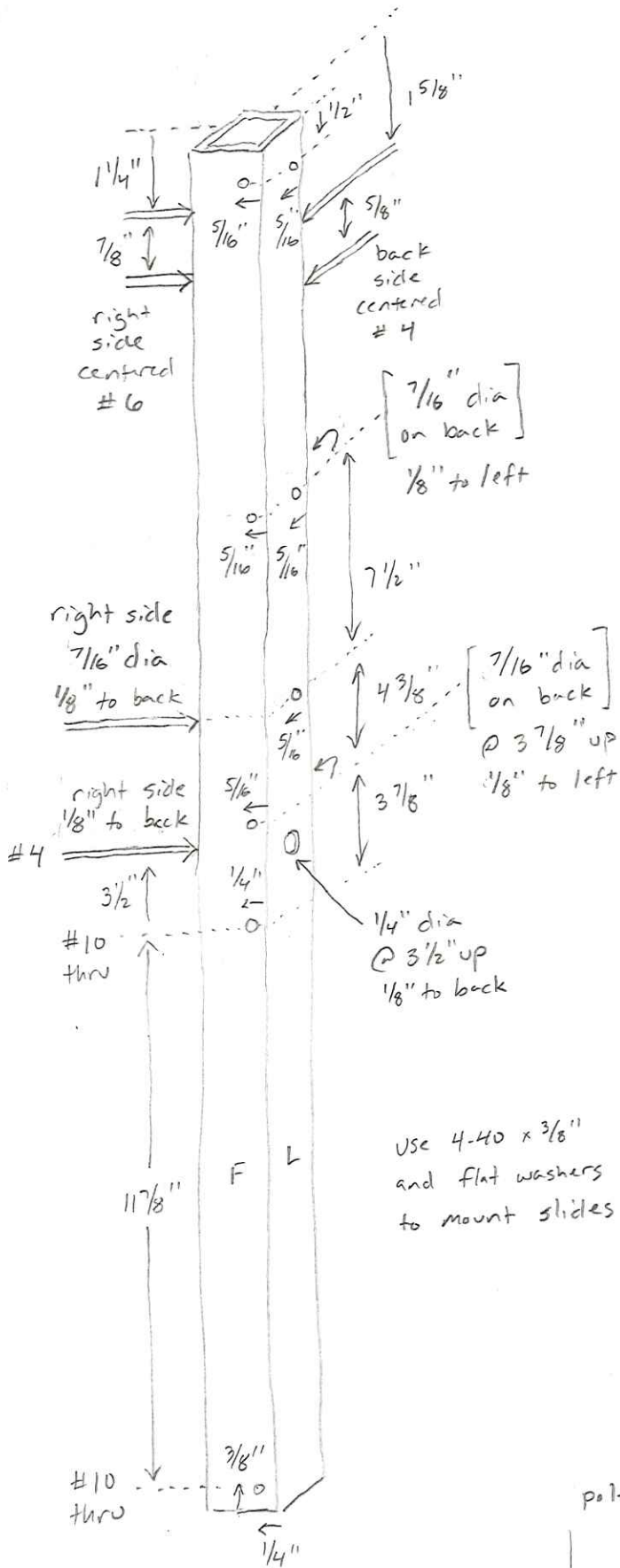




electronics tray
12" x 6 3/8" overall
0.040" aluminum

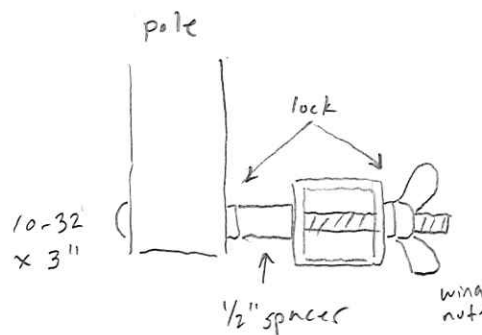
folds in 0.051"
very difficult —
use 8 7/8" x 1/2" L
in front instead

Eli shelf
+ tray
JHC 10/19/11



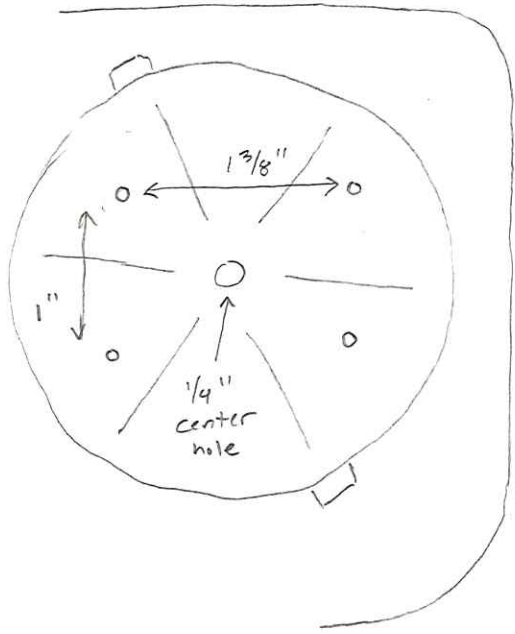
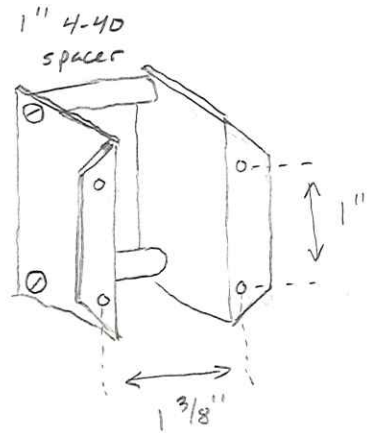
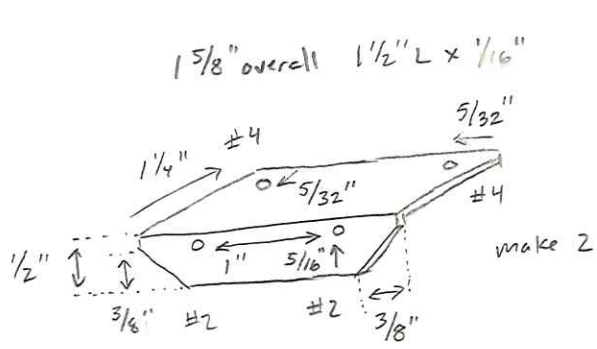
Use 4-40 x 3/8"
 and flat washers
 to mount slides

40 3/8" overall
 1" tube x 1/16"
 pole

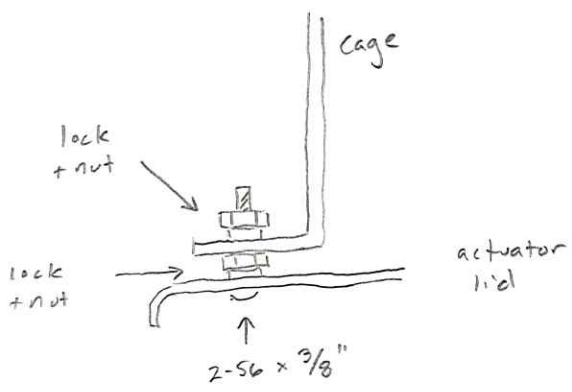


Eli pole + brackets

JHC 10/27/11

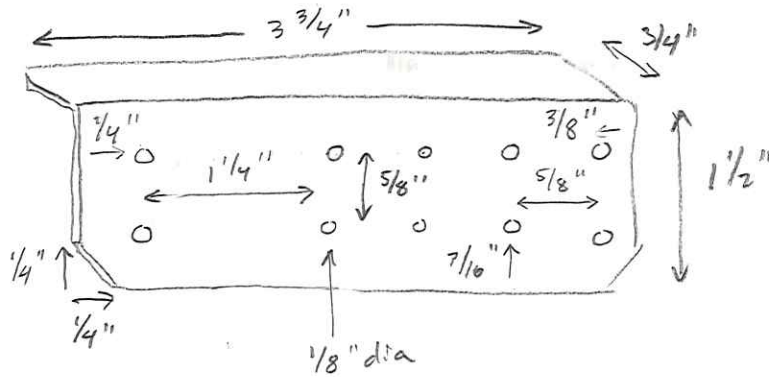


4 holes aligned with vertical, not spokes drill for #2

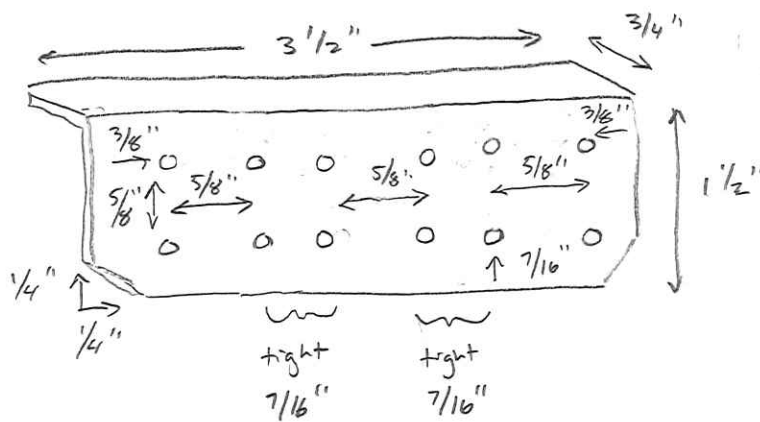


Eli Lift Pot Cage
JHC 11/29/12

OBSOLETE



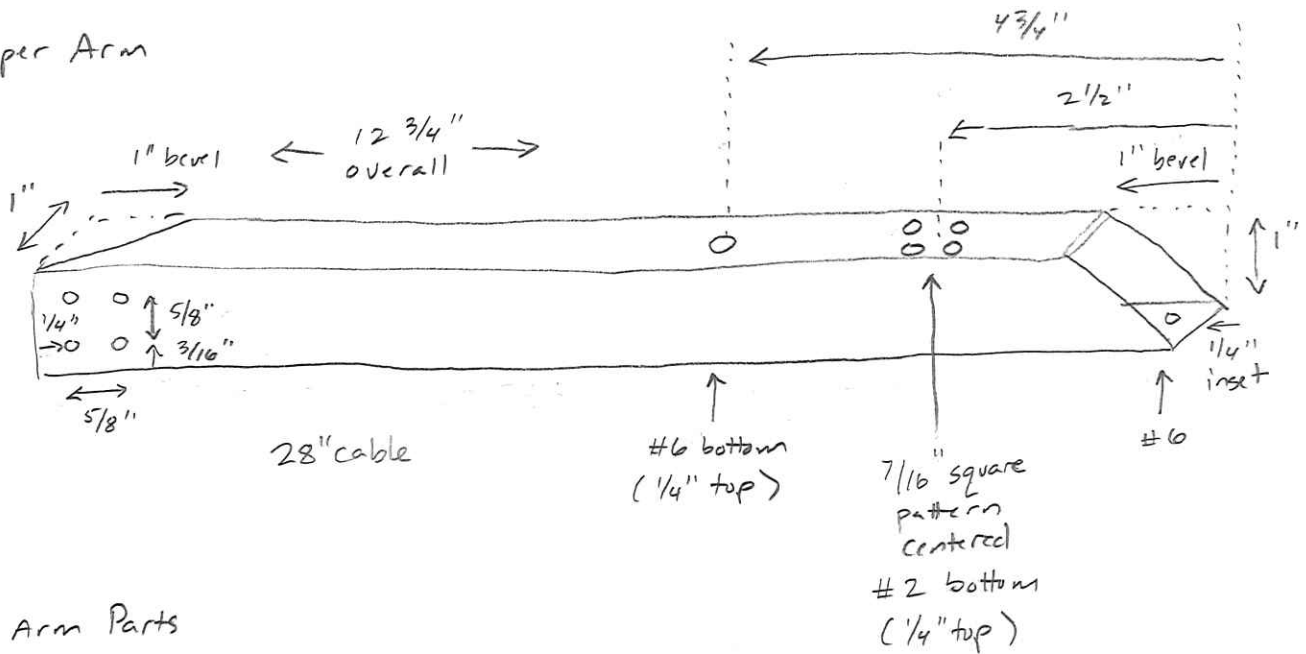
Lift
Bracket



Elbow
Bracket

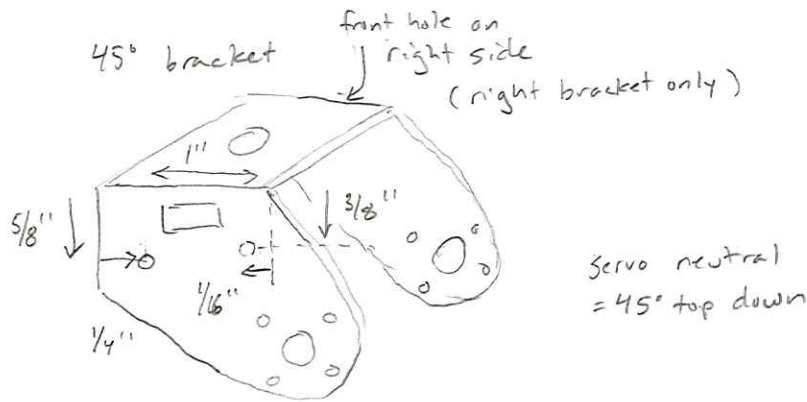
use Loctite
no washer
on middle

Upper Arm

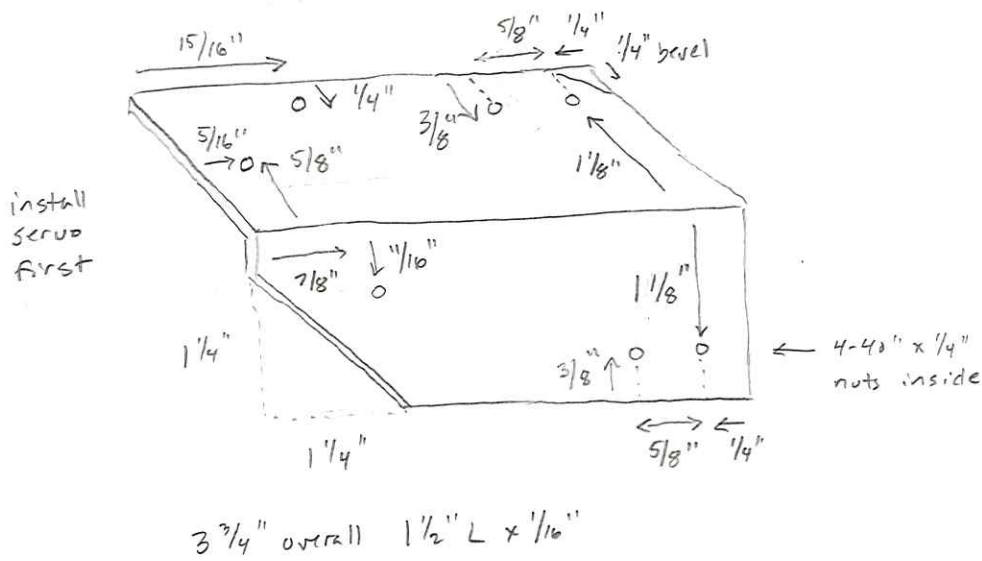


Elbow Arm Parts

JHC 5/4/11



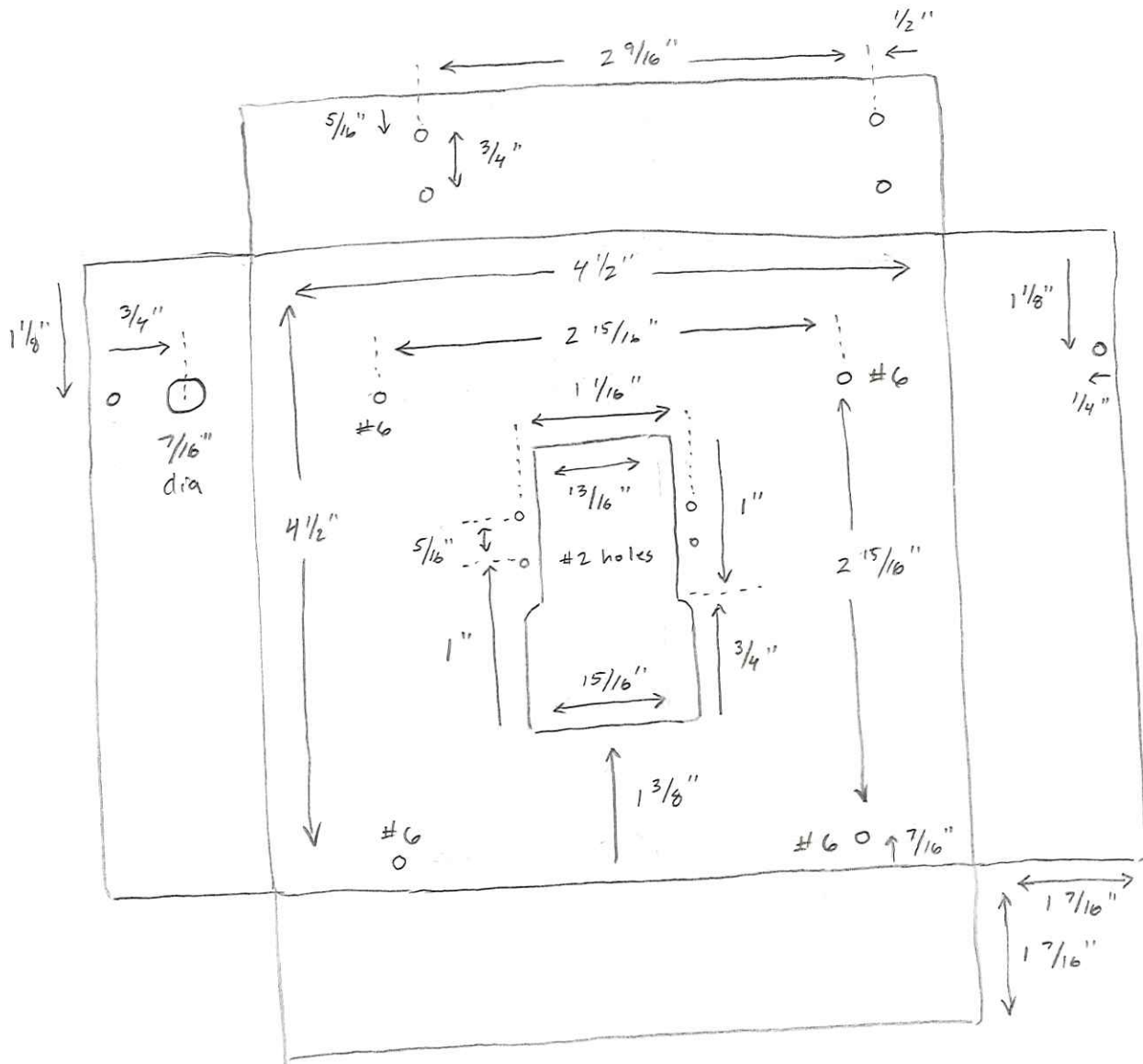
use 2x 3/8" spacers
make 2 symmetric (R shown)



Modified Eli Arm Lift
JHC 9/21/11

$7\frac{3}{8}$ " square $\times$ $\frac{1}{16}$ " overall

Top View - Bend on OTHER side

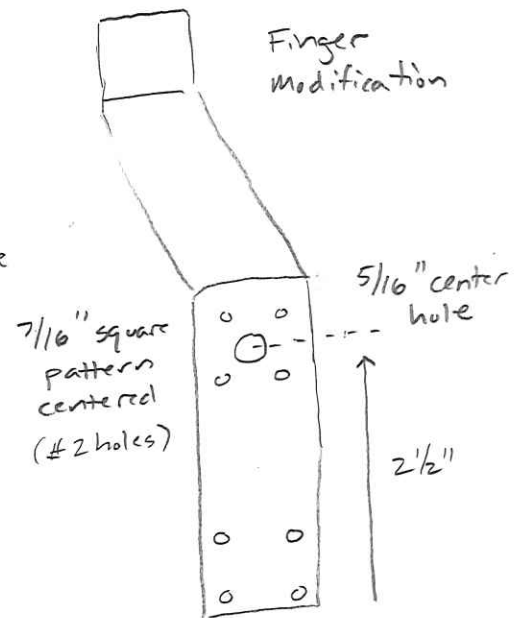
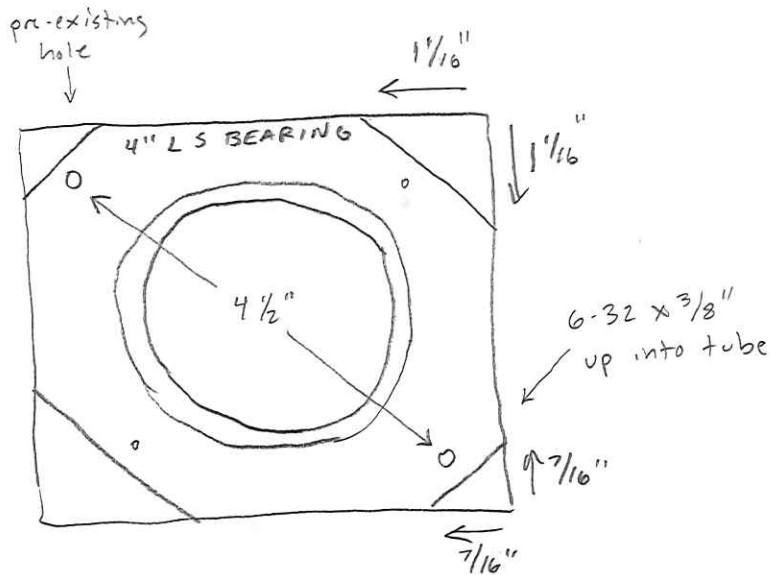


machine fold sides
crease front + back

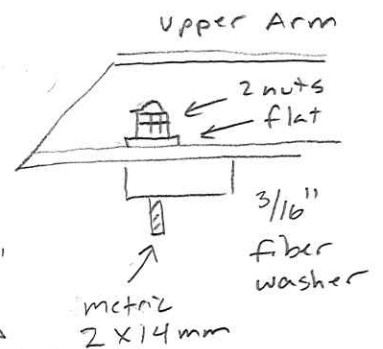
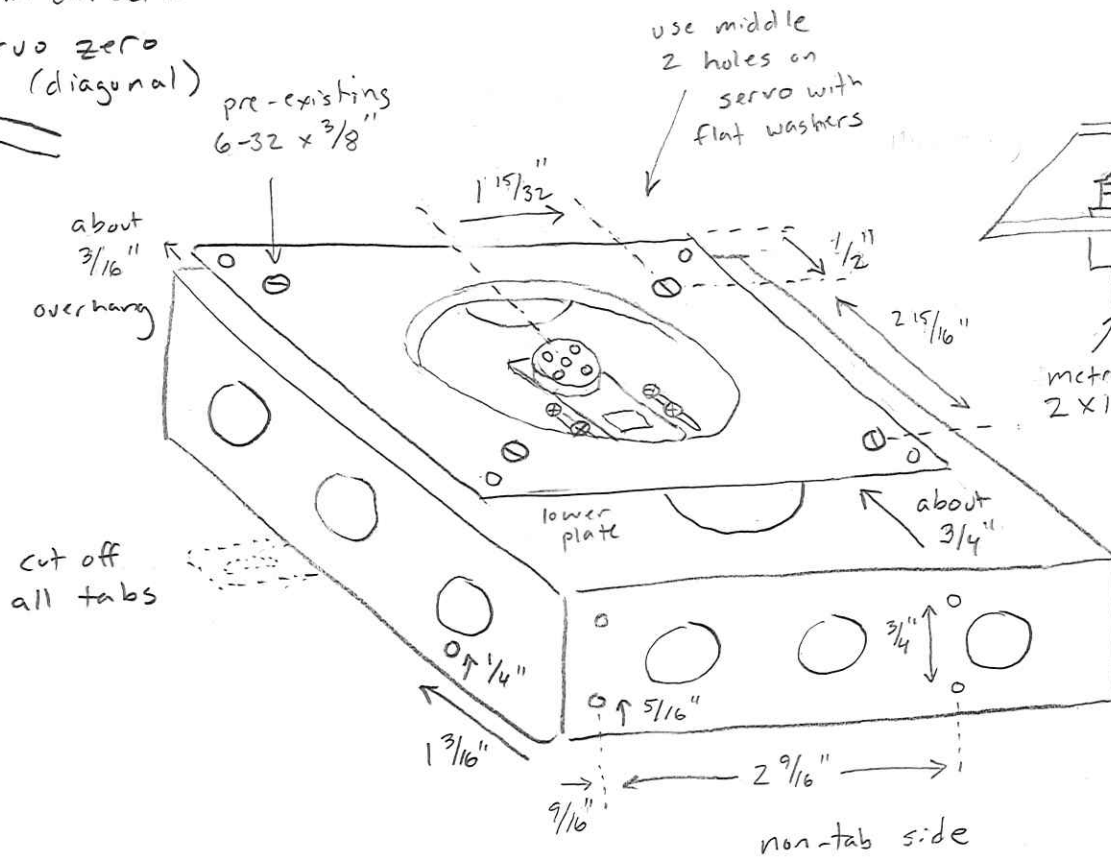
El: Substitute Arm Base

JHC 1/1/13

Lazy
Susan
Trimming



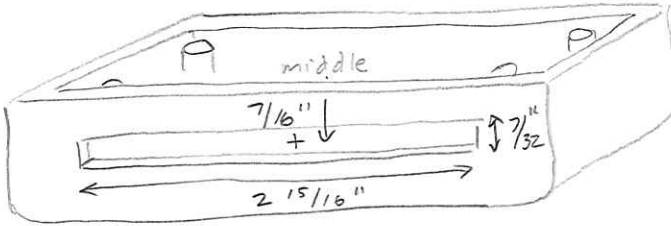
upper arm direction
for servo zero
(diagonal)



Eli Arm Base

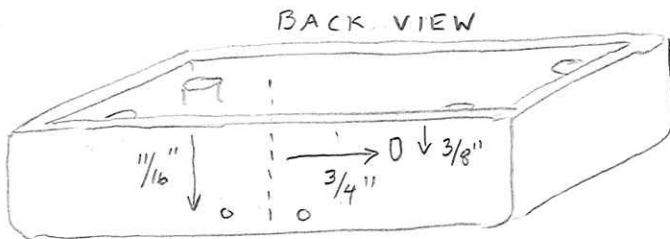
JHC 5/4/11

Alternatively
3x1\"/>

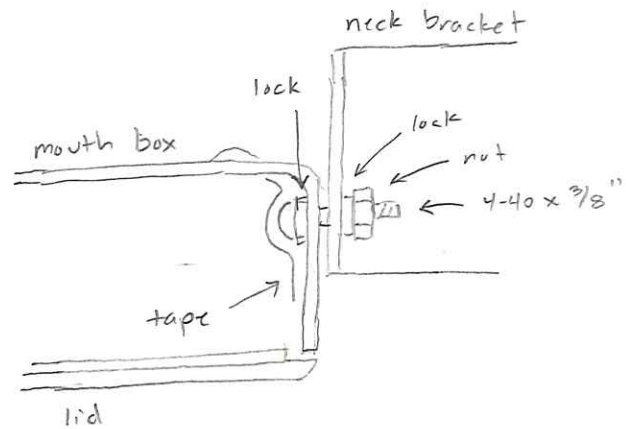


rough out with
Dremel cutoff
wheel on low

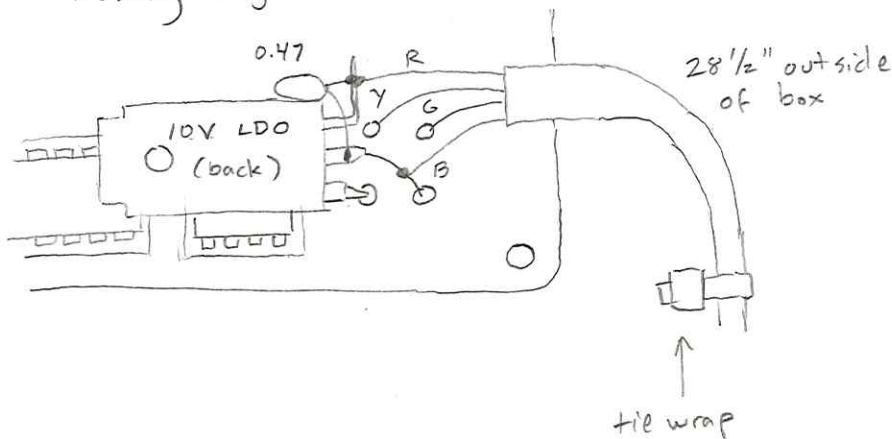
FRONT VIEW



#4
 $\frac{5}{8}"$

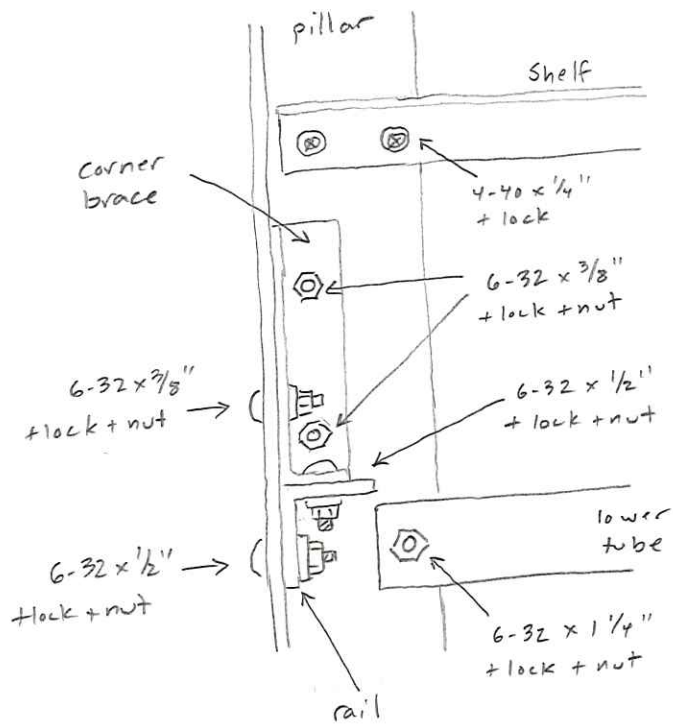


Adding Regulator

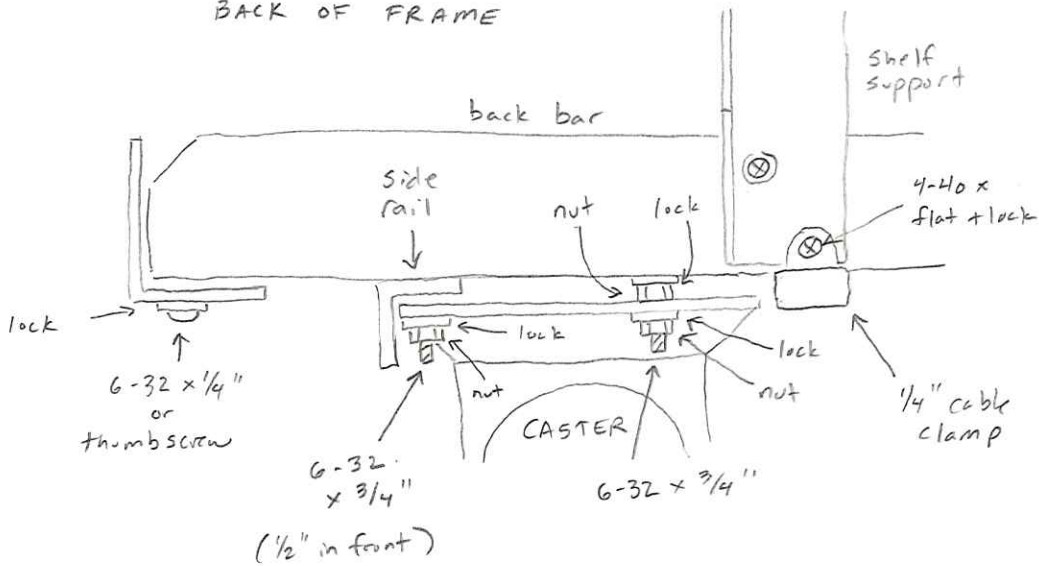


Eli Mouth Box
JHC 11/25/11

FRONT OF FRAME



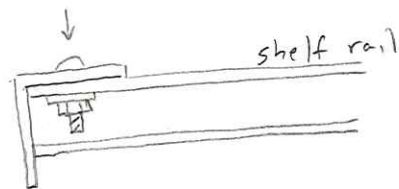
BACK OF FRAME



E1: Frame Assembly

JHC 3/4/13

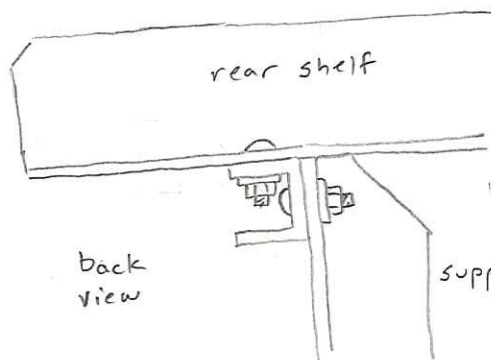
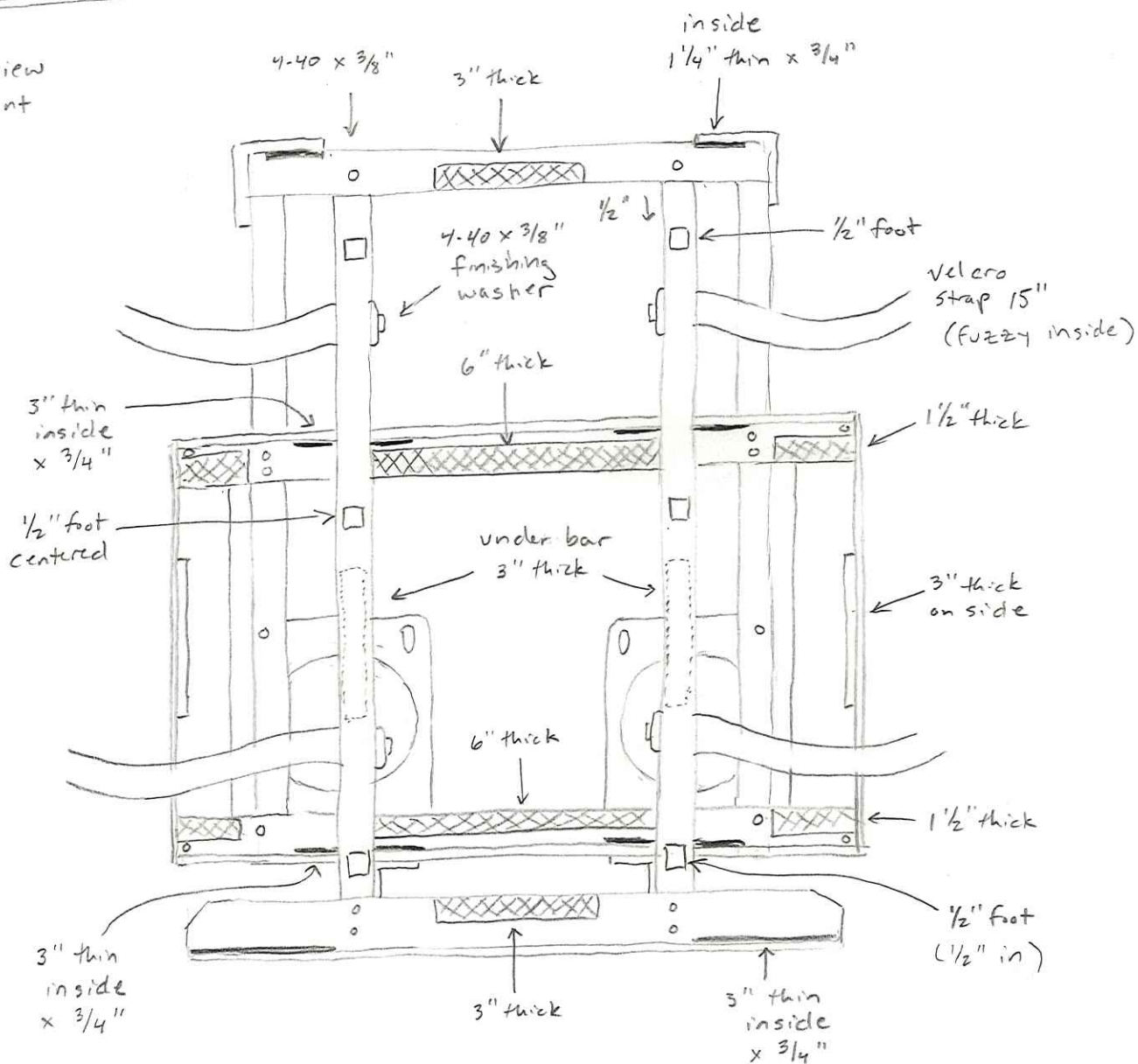
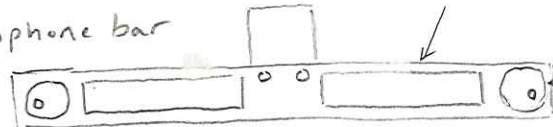
4-40 x 3/8"



side view
of front

microphone bar

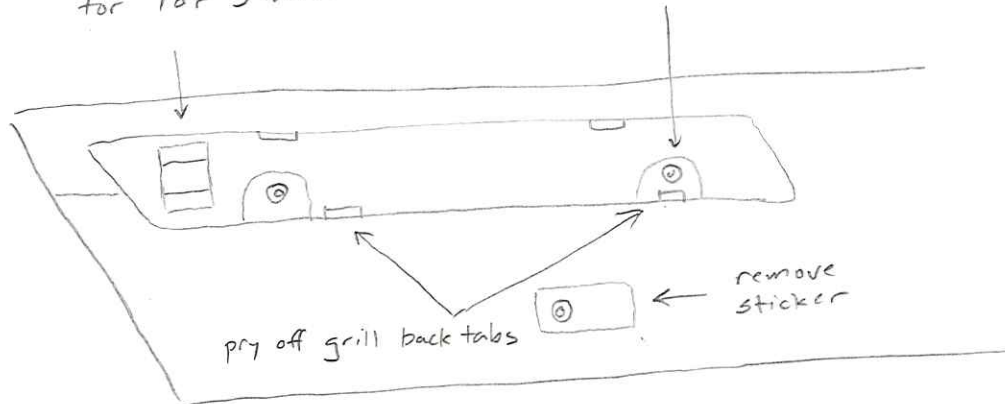
2 1/2" thin x 5/8"



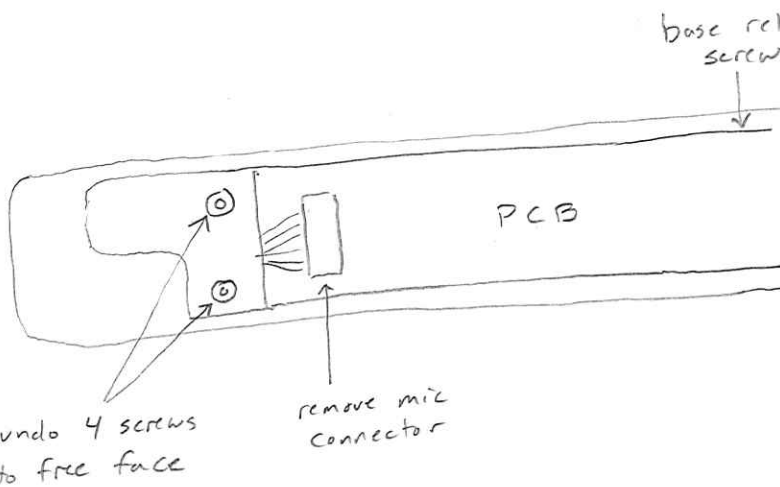
Eli Paddling
JHC 3/14/12

grills have patterns for 1 or 3 mics

use offset $\frac{3}{32}$ " flat head or break central post and use Torx head

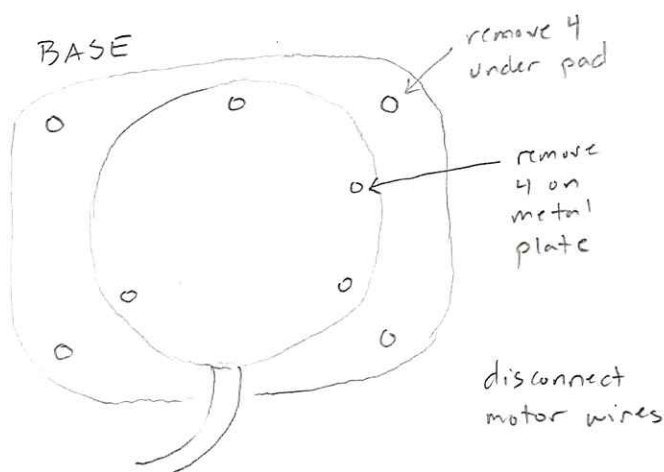
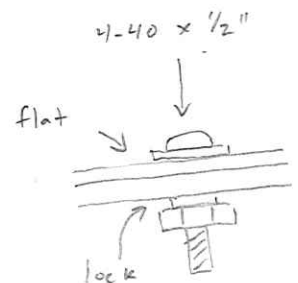
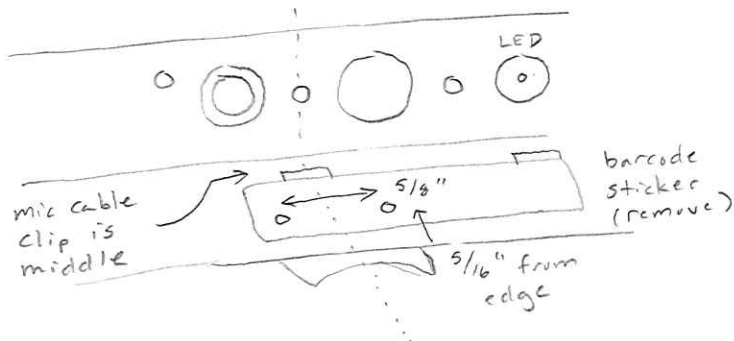


remove 6 screws then take off top + bottom shells and side grills (fan)



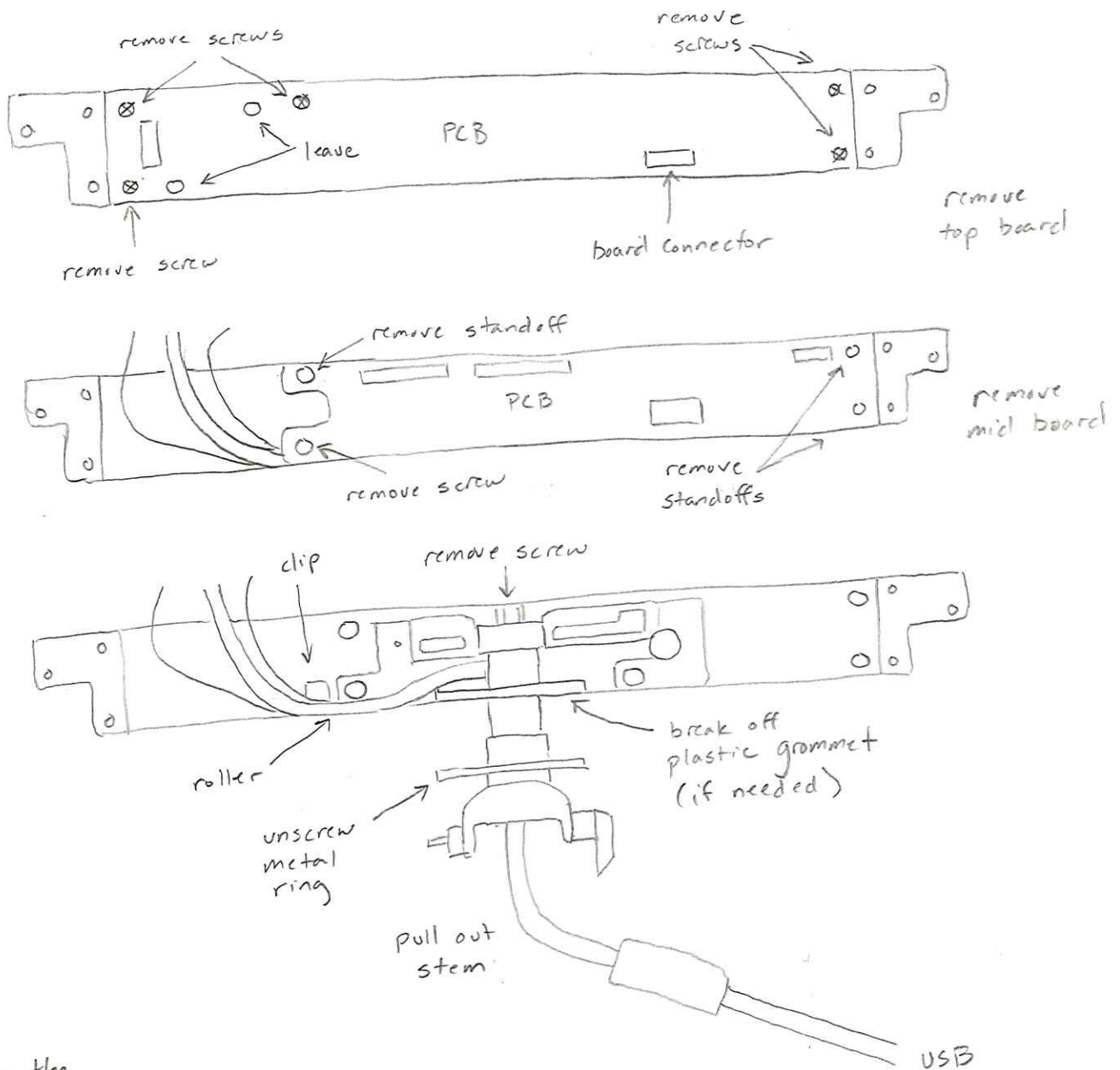
remove face + mic plates for drilling + release LED

Drill mic plate then mark + drill face plate

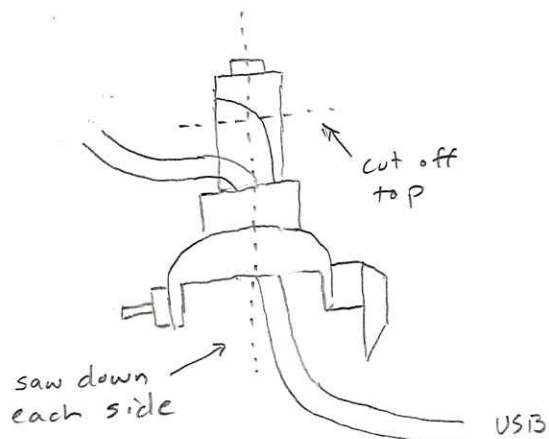
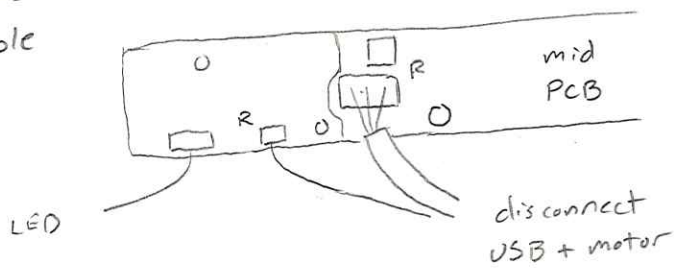


Eli Kinect DisAssembly
JHC 1/31/13

about 60 min

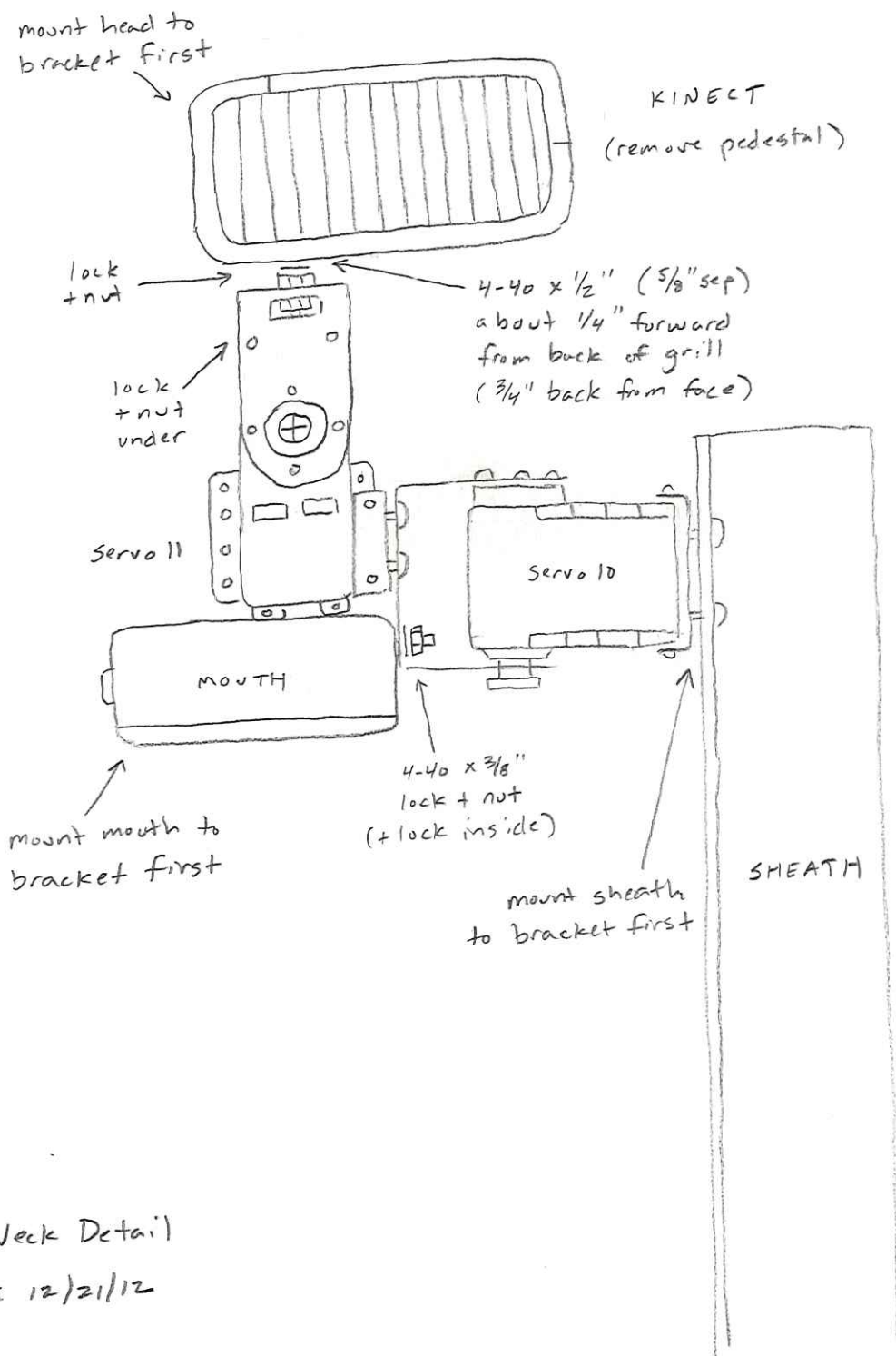


Freeing the
USB cable



Eli Neck Removal

JHC 2/1/13

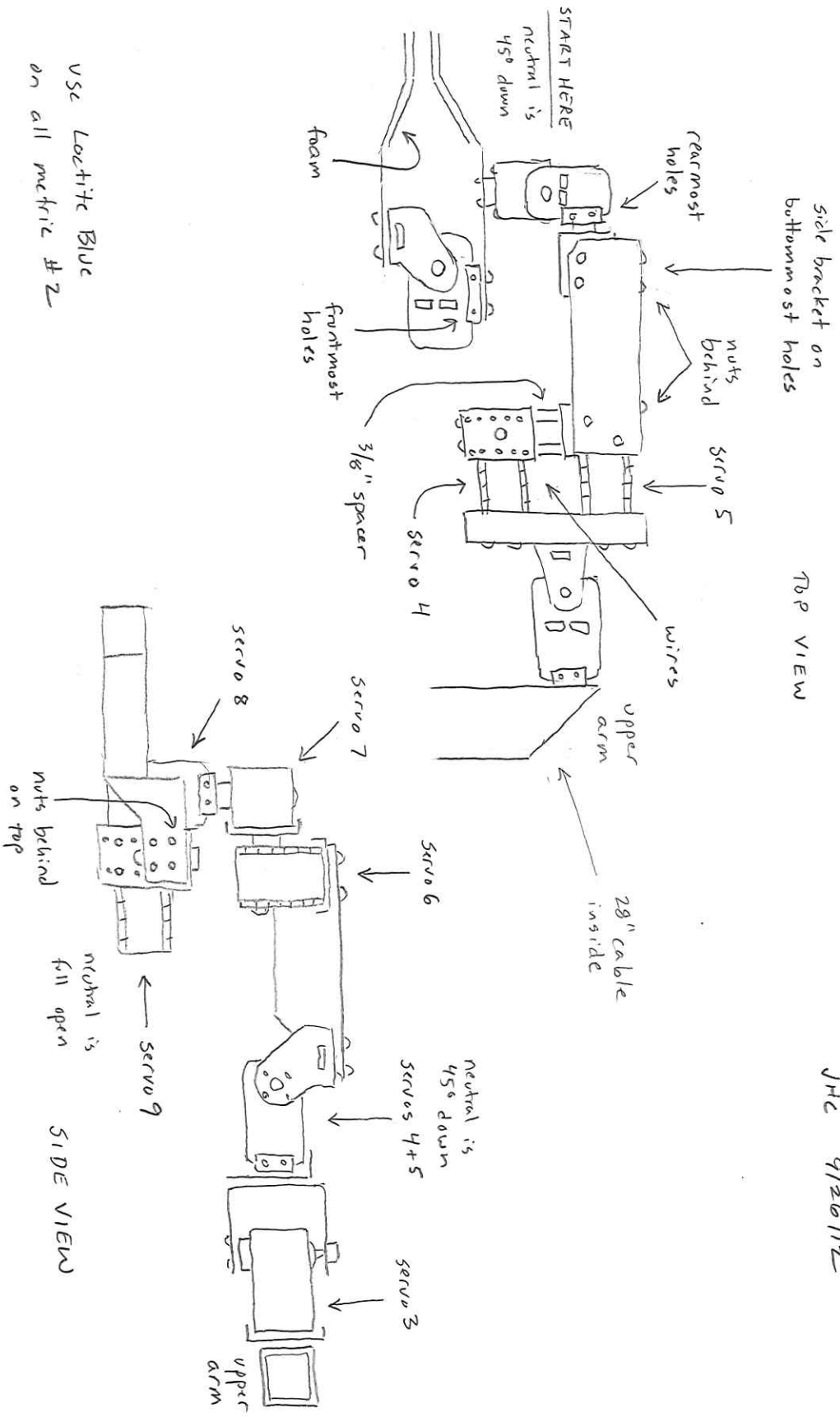


Eli Neck Detail

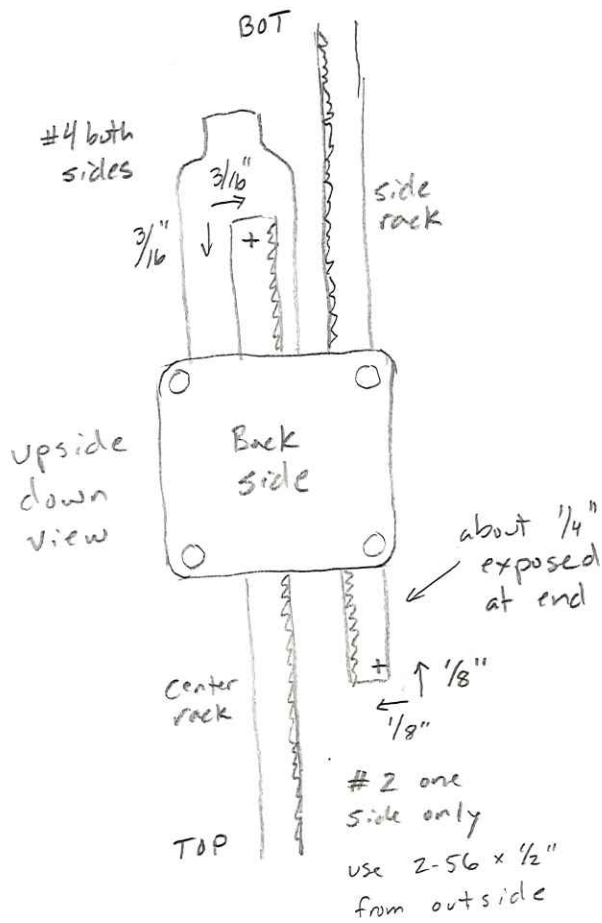
JHC 12/21/12

Eli Lower Arm Assembly

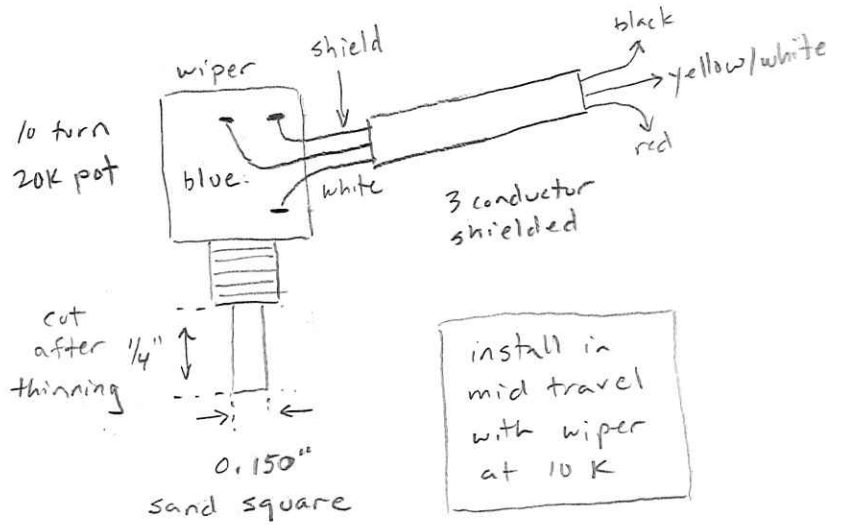
JHC 9/26/12



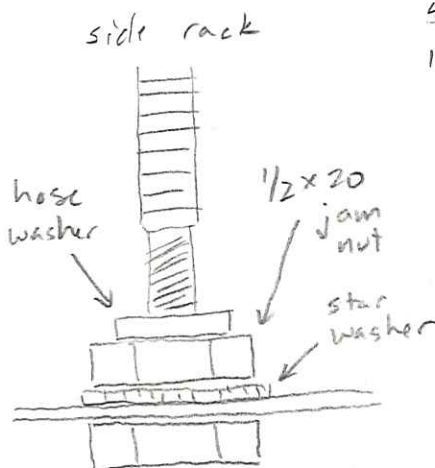
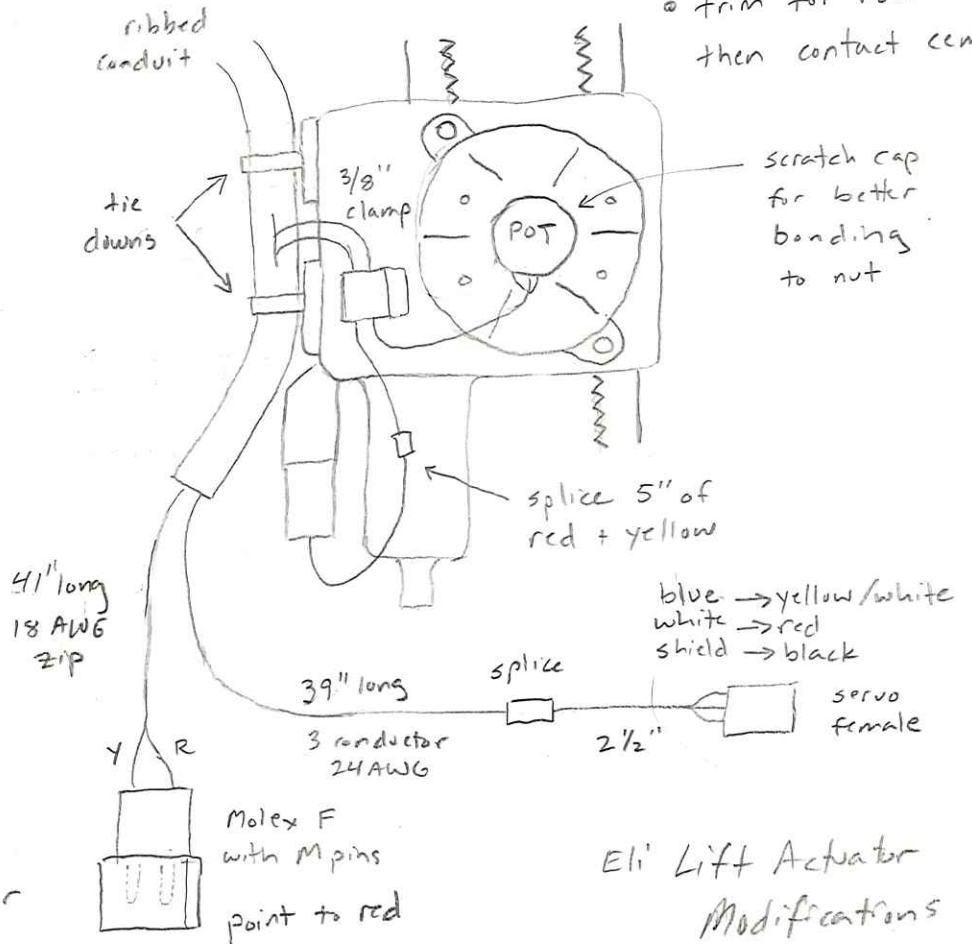
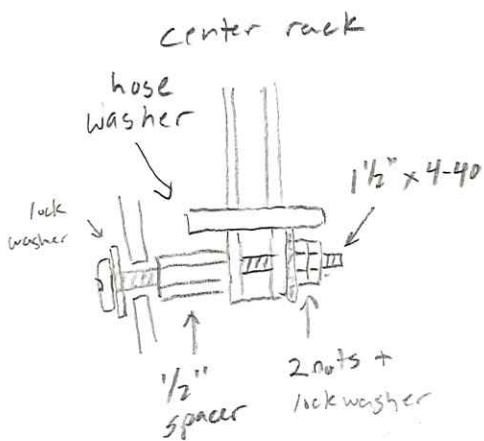
Use Loctite Blue on all metric #2



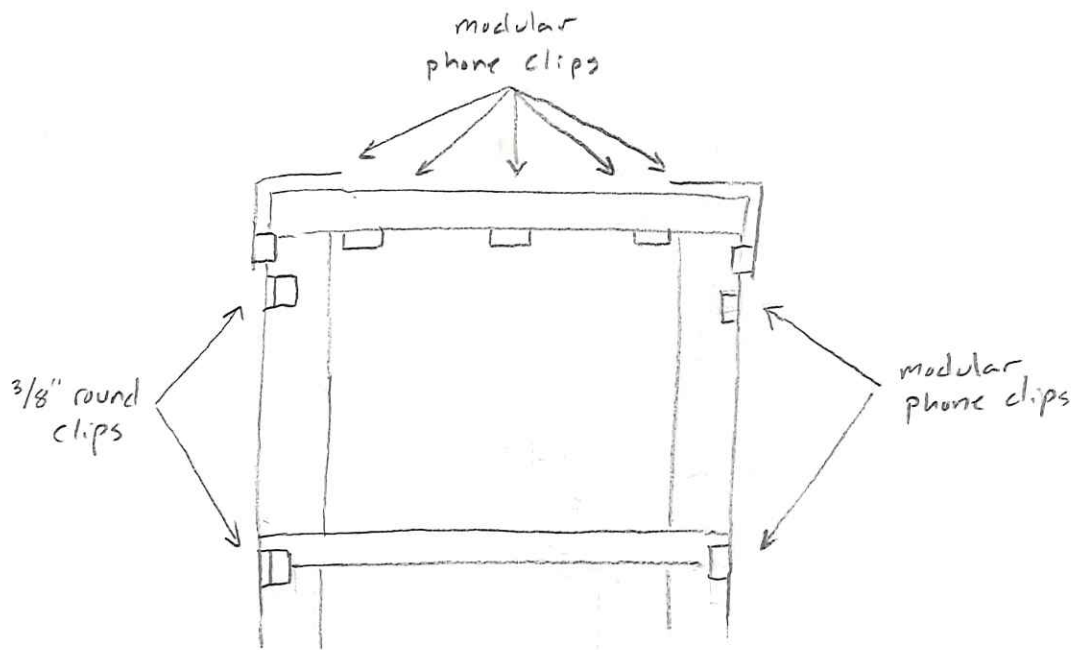
roughly 6-14K range



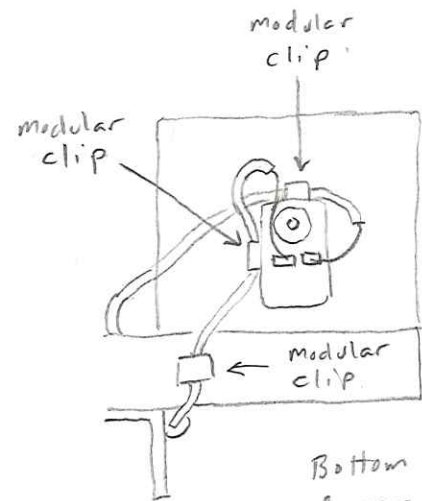
- Drill $1/4"$ hole in pinion cap
- Rough up hole in pinion shaft
- Hot melt glue pot bushing + nut to cap
- Hot melt glue pot shaft in hole
- trim for rotation then contact cement



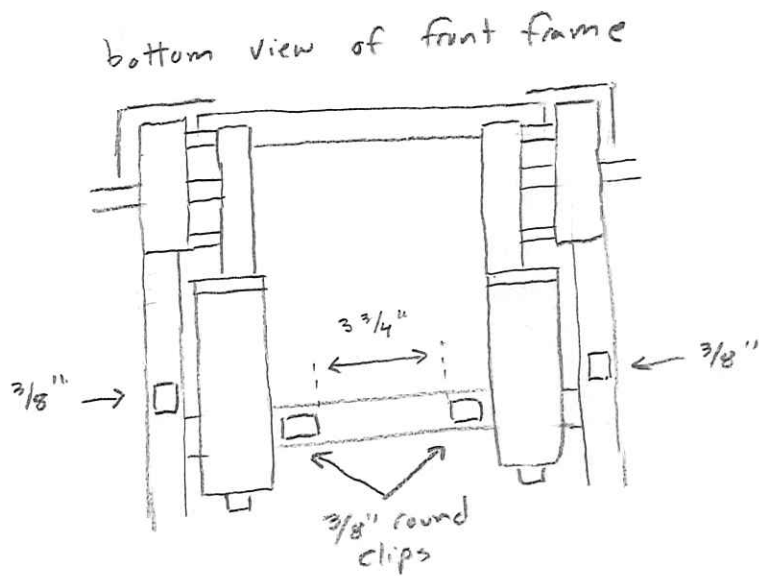
Eli's Lift Actuator Modifications
JHC 4/22/11



rear view of front frame

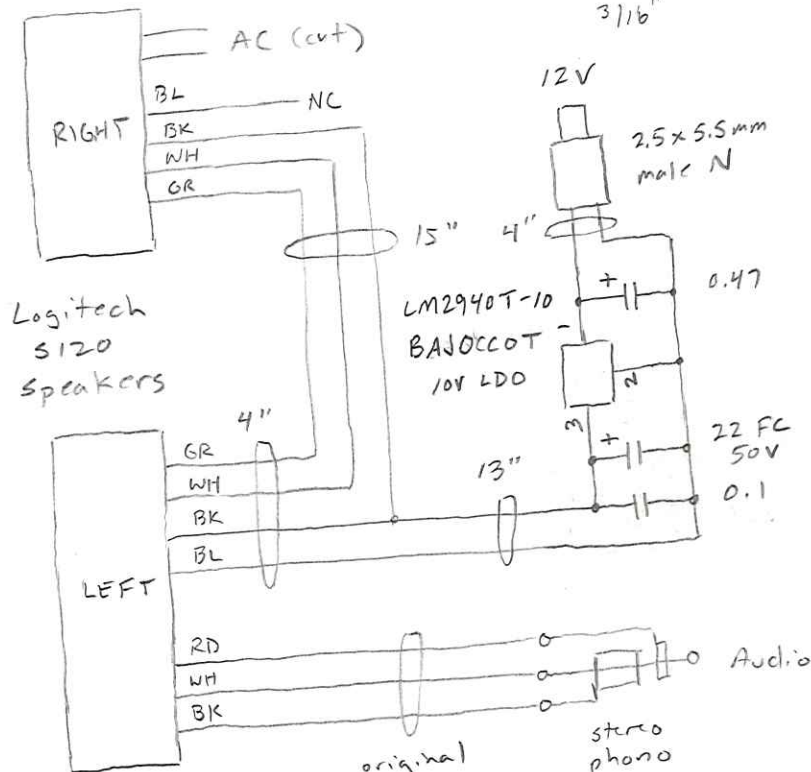
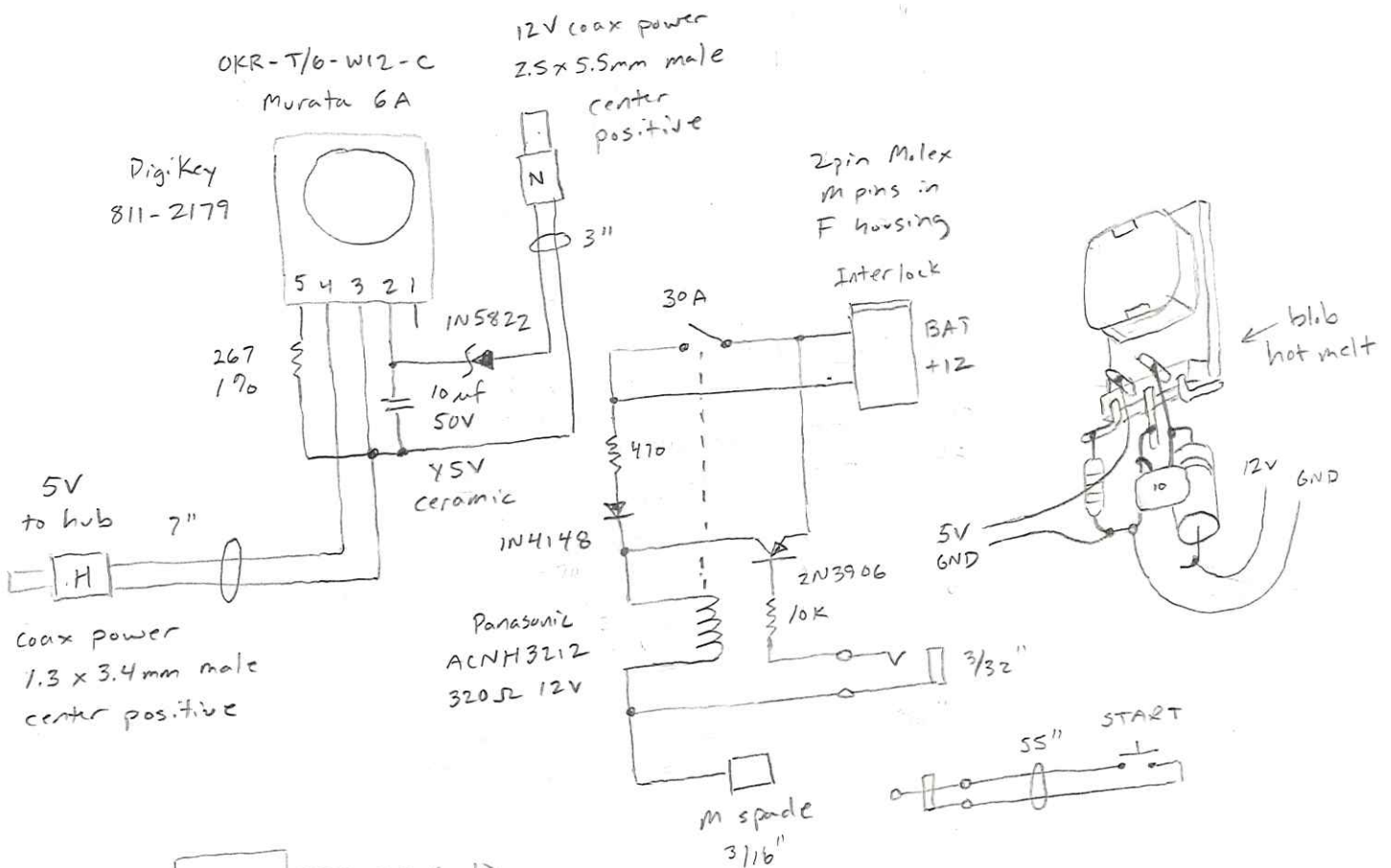


Bottom view of arm base

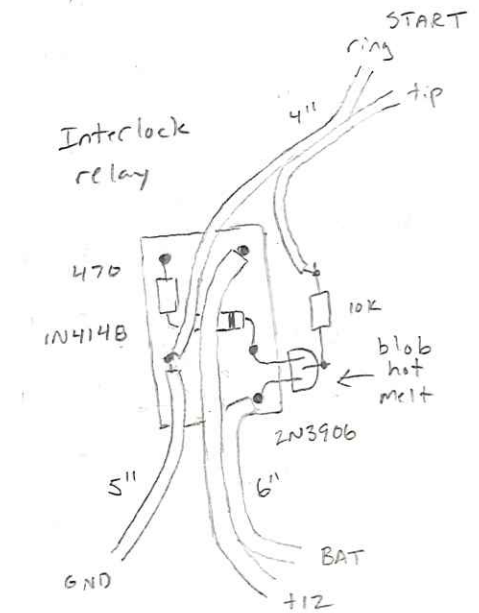
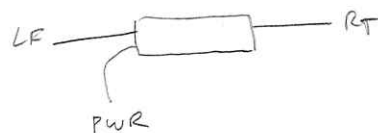


Eli Cable Clamps

JHC 1/4/13



- tail overhangs
by 1 1/4"
- mount with
foam tape

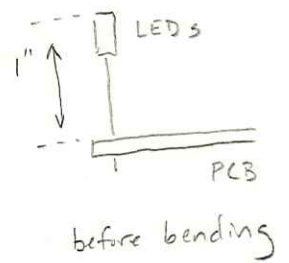
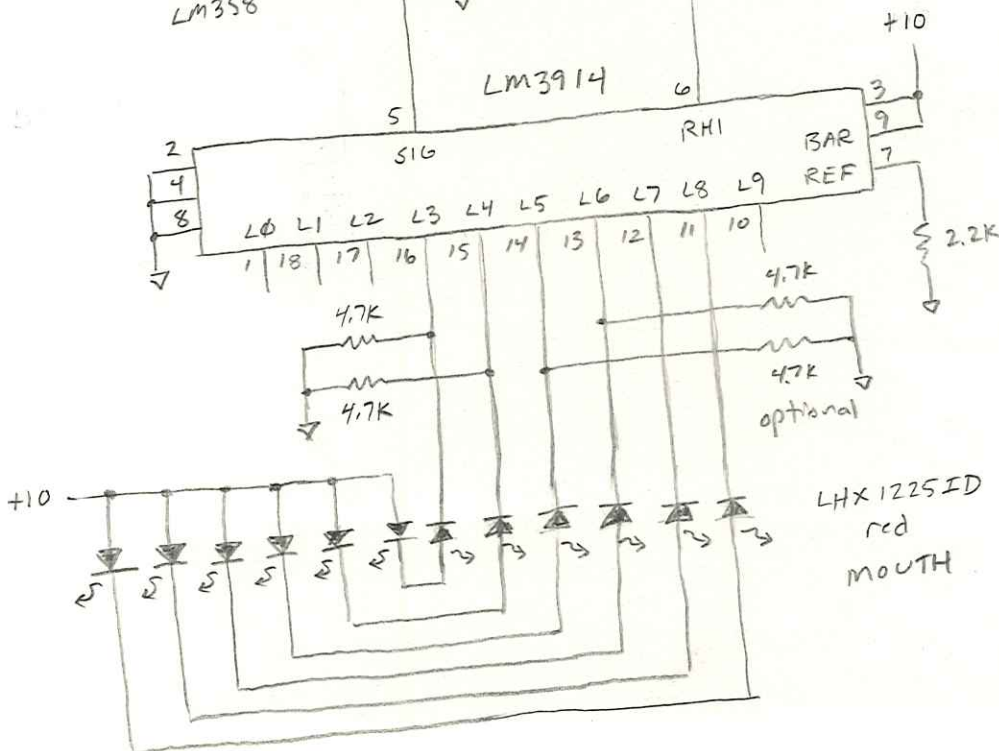
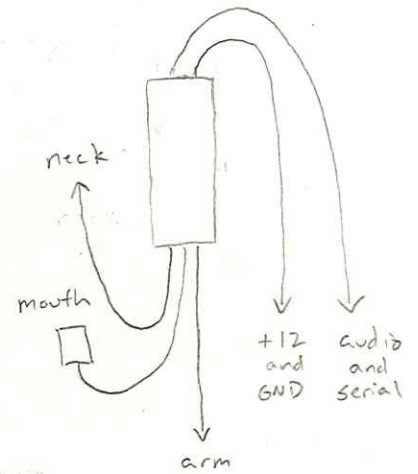
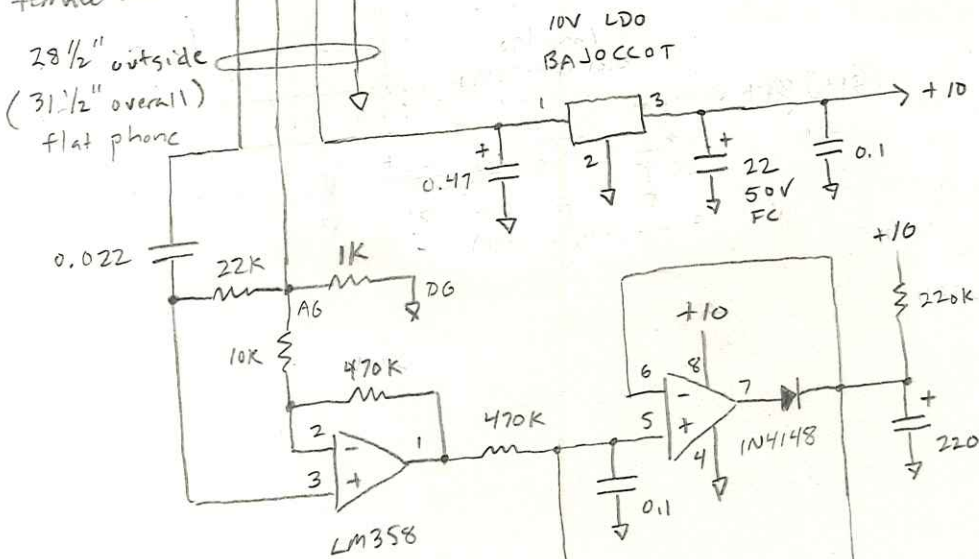
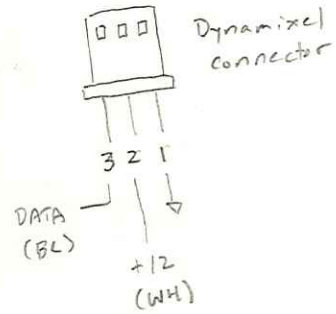
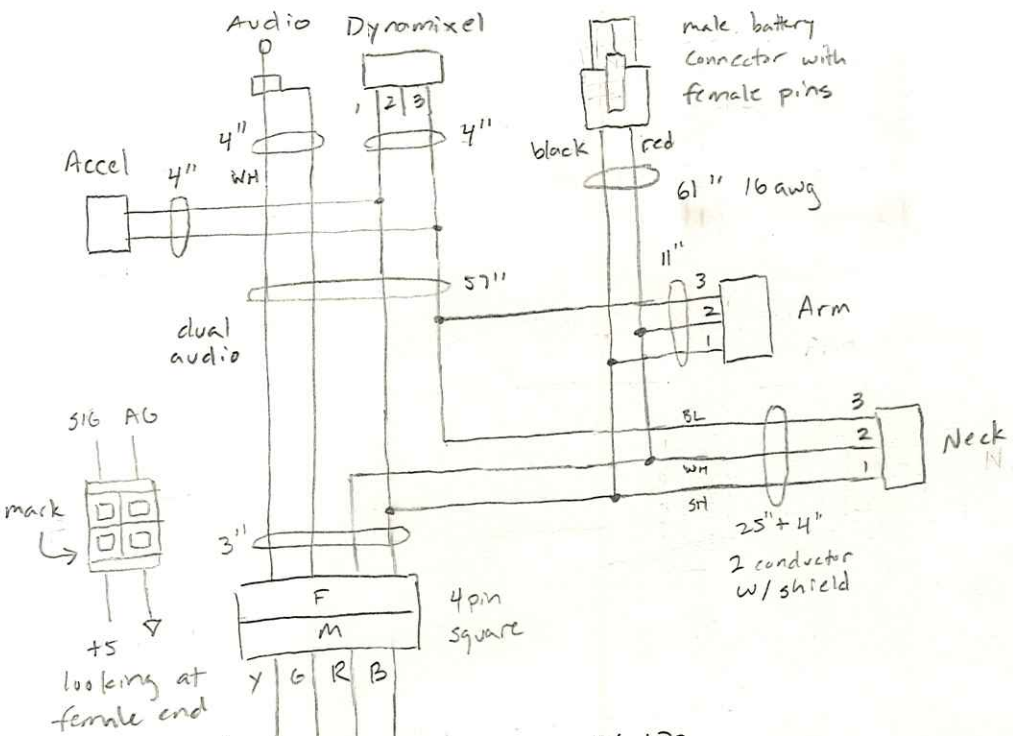


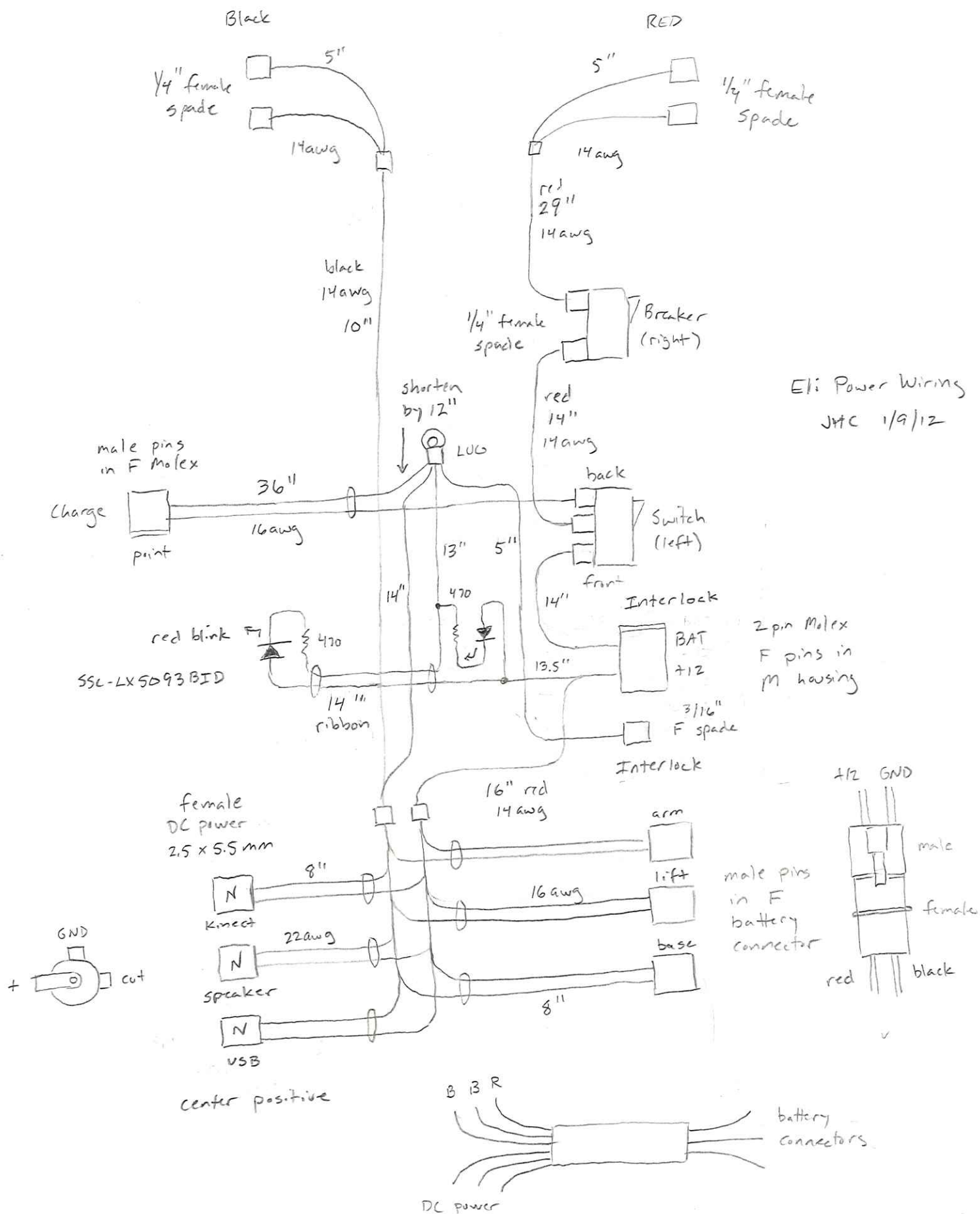
Eli Power Converter
+ Speaker Wiring

JHC 1/25/12

Eli Head + Wiring

JHC 12/20/11

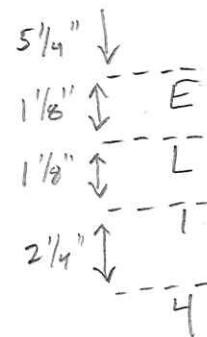
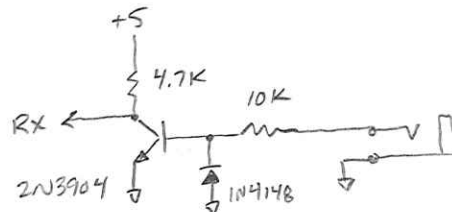




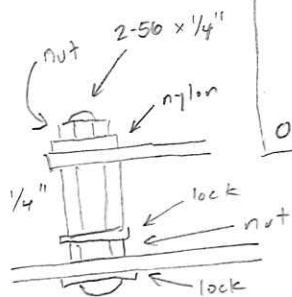
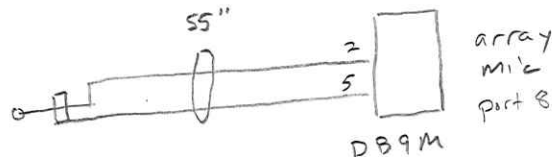
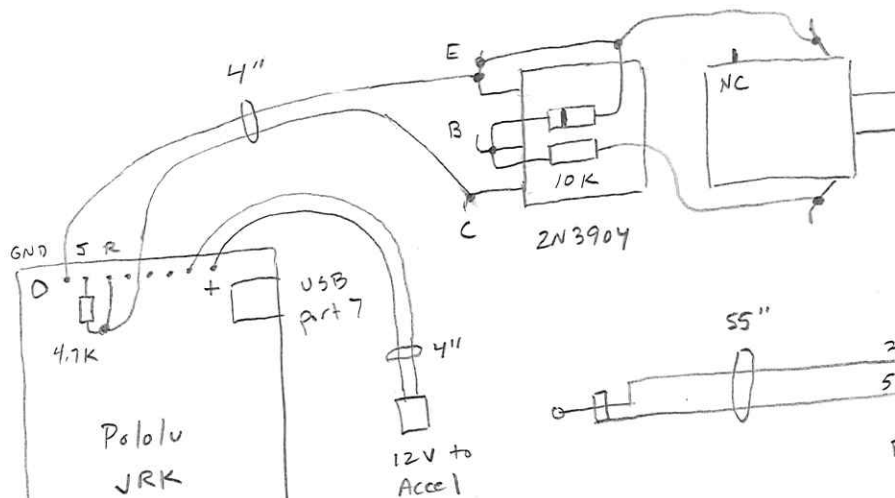
Labels:

IBM ← 3/4" tape Auto, Brussels

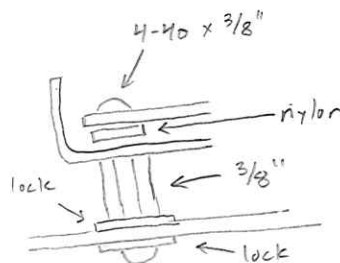
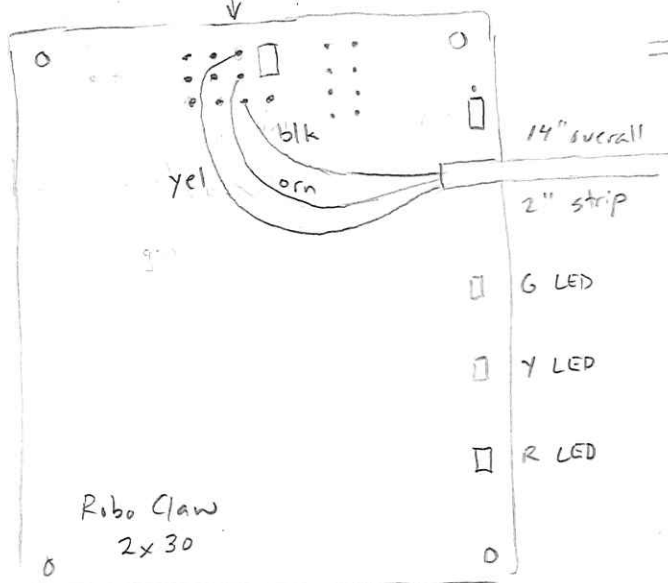
CHARGE RUN } Helsinki
BATTERY 12 pt
CHARGE 1/2" tape
START ← 18 pt
BOLD Helsinki



1" vinyl letters



can use 3 pin header here

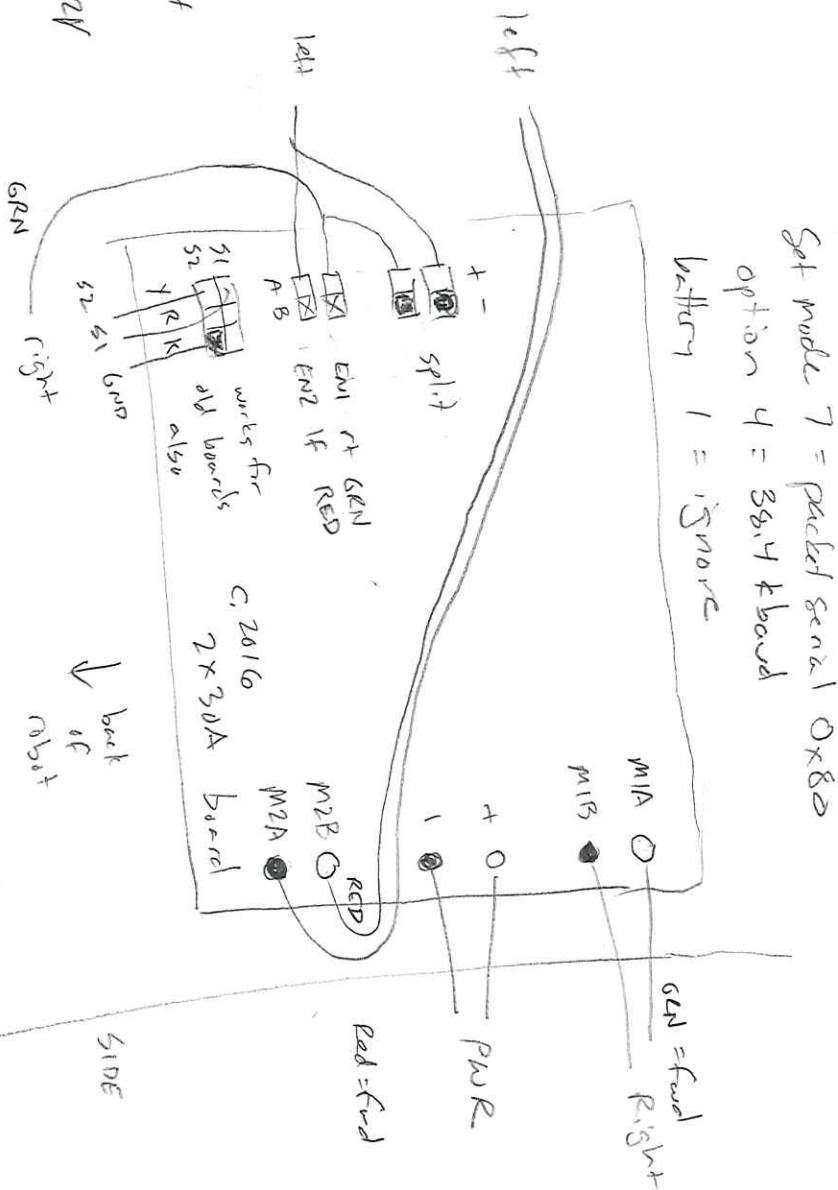
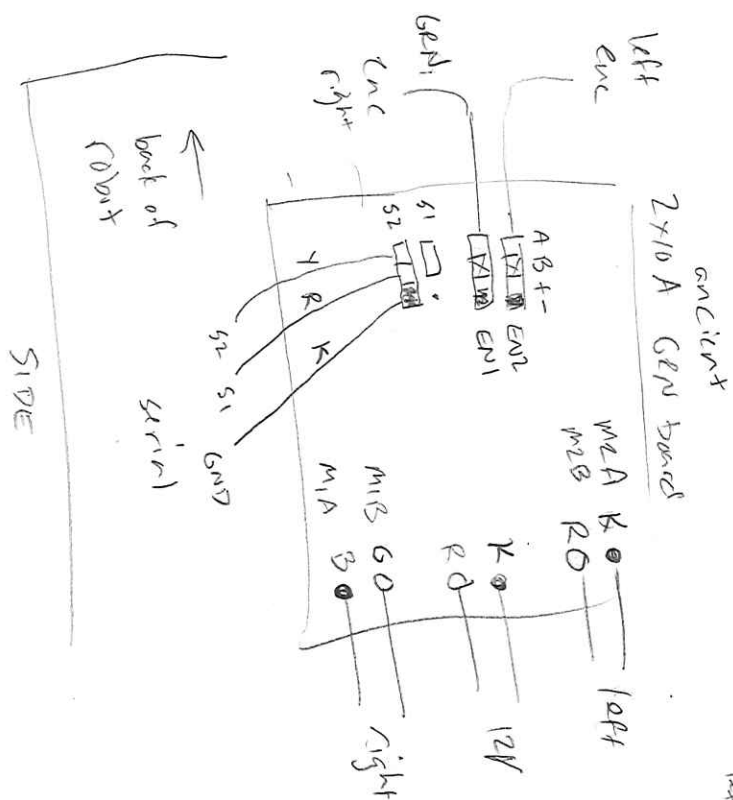


FTDI
USB-RS232
SV WE
part 6

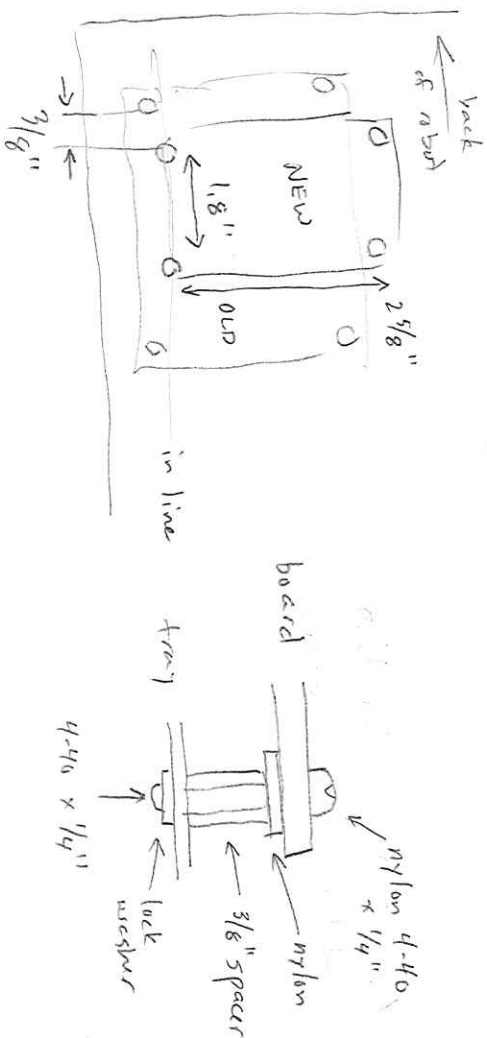
Eli Controller Communication
JMC 5/16/12

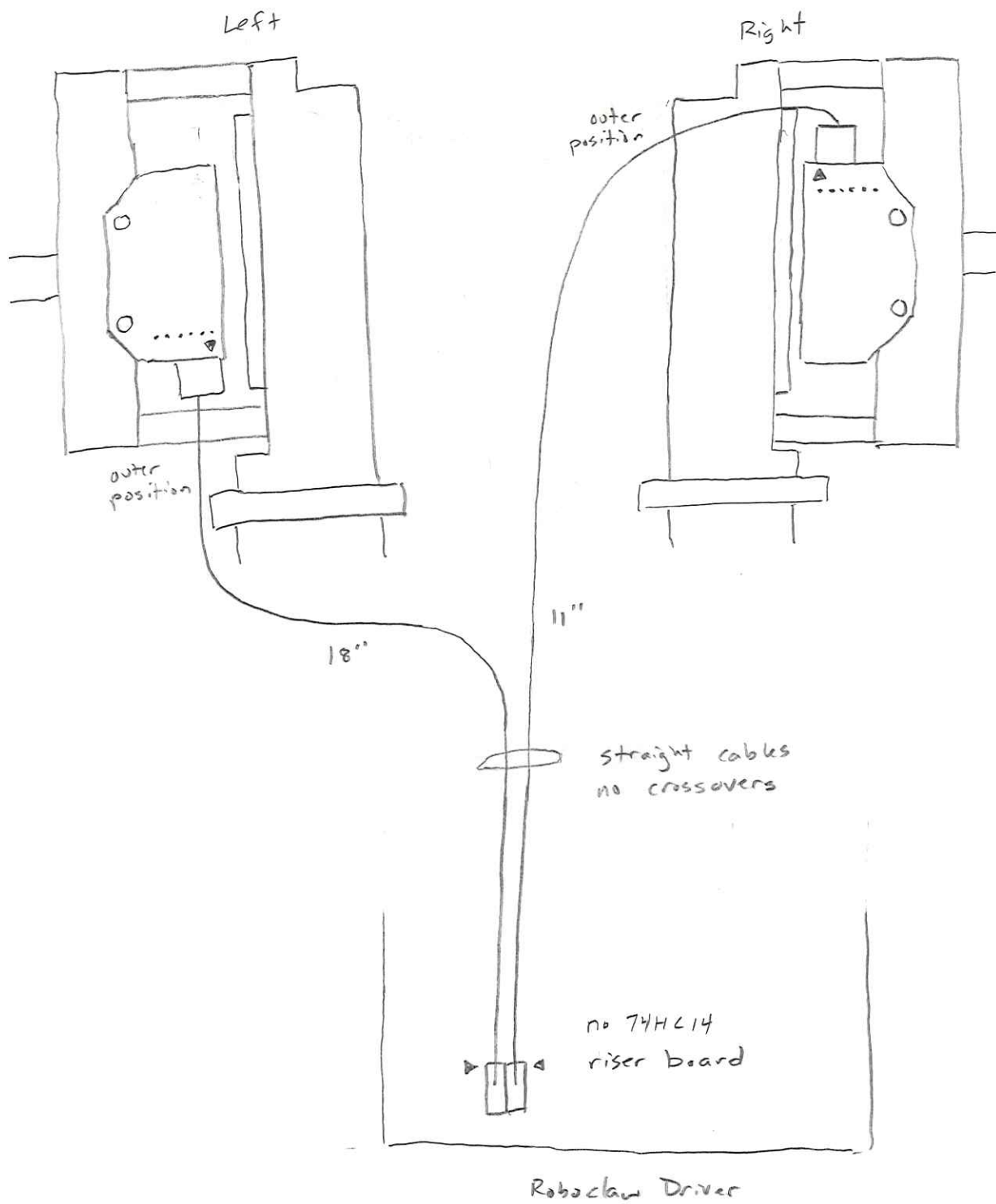
2010 Base Controller Changes

4/11/10



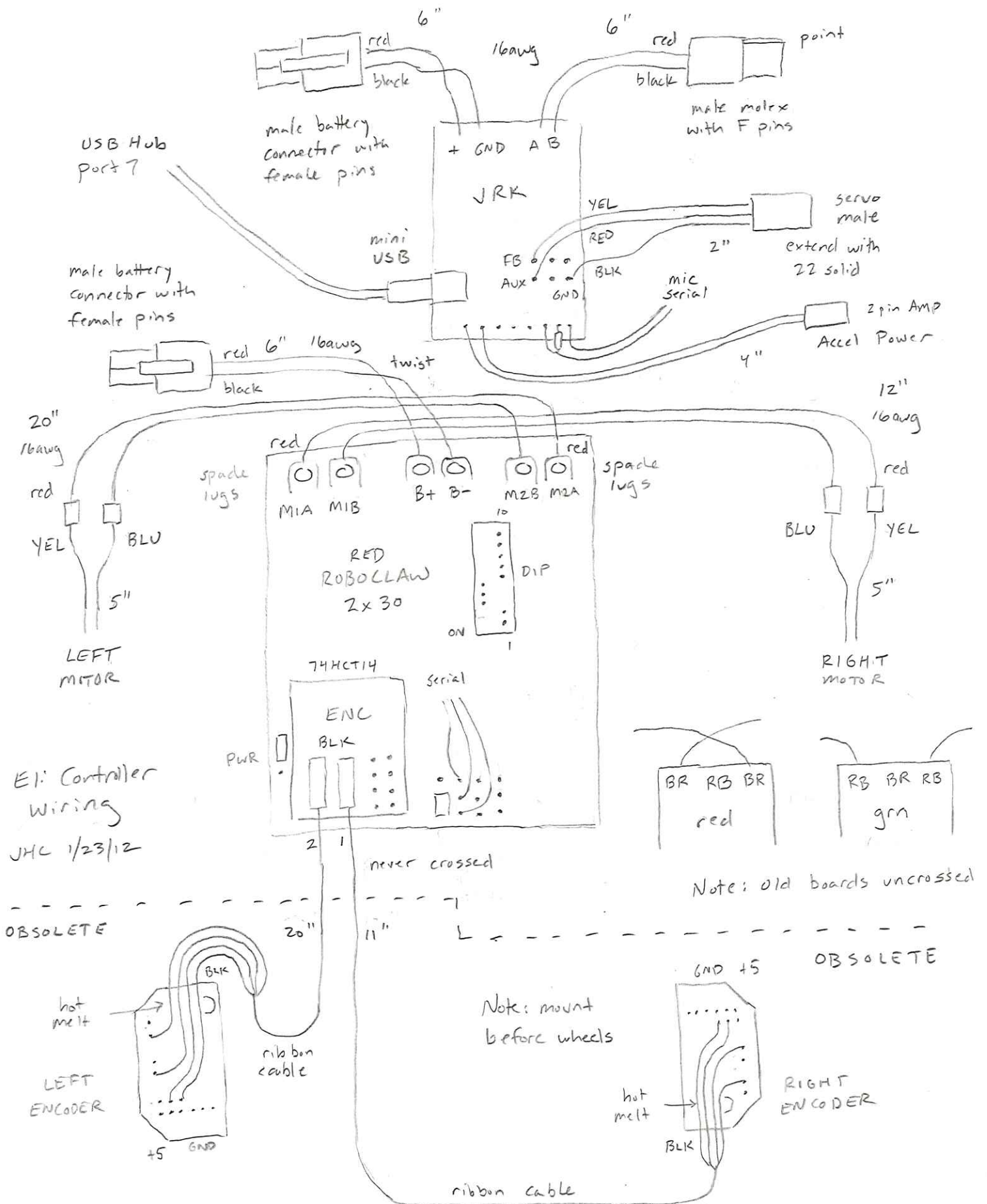
Set mode 7 = packet serial 0x80
option 4 = 38.4 kbaud
battery 1 = ignore

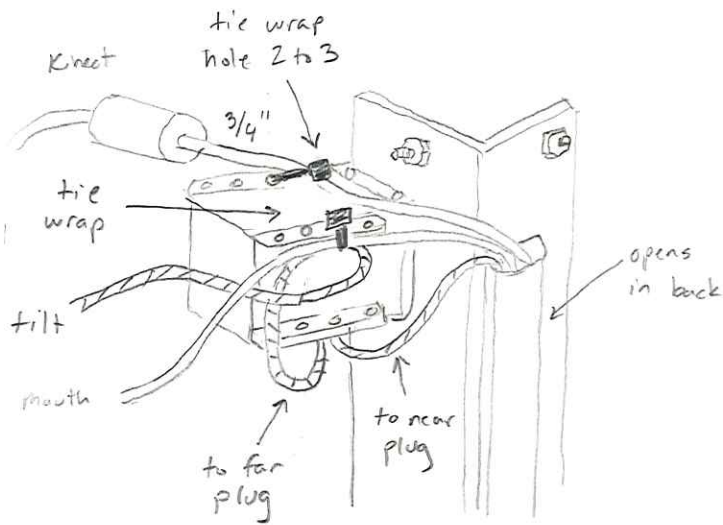




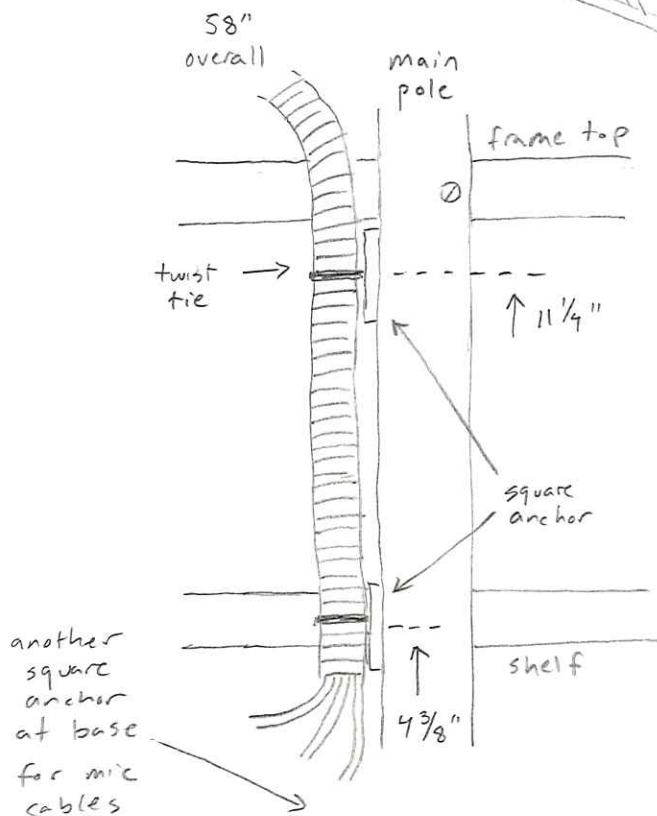
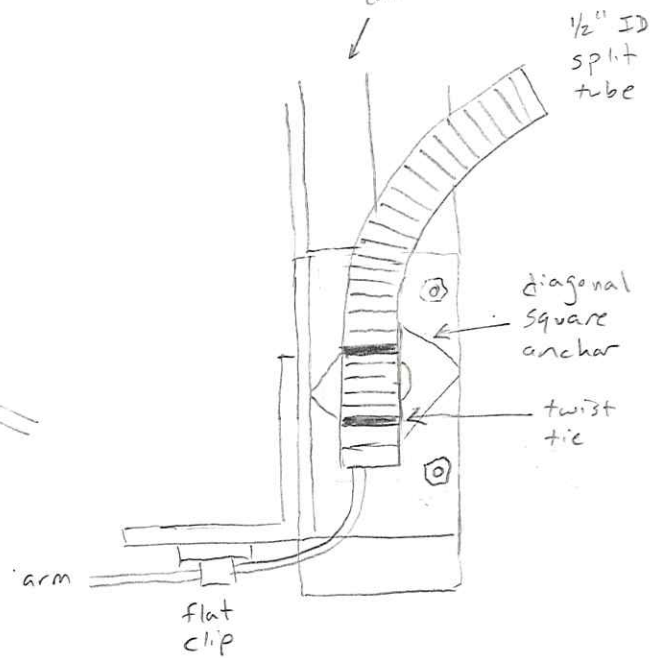
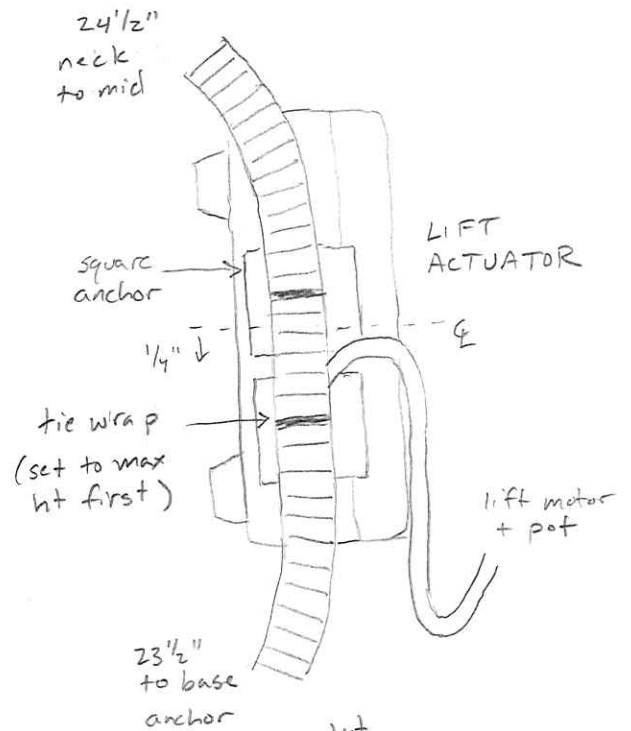
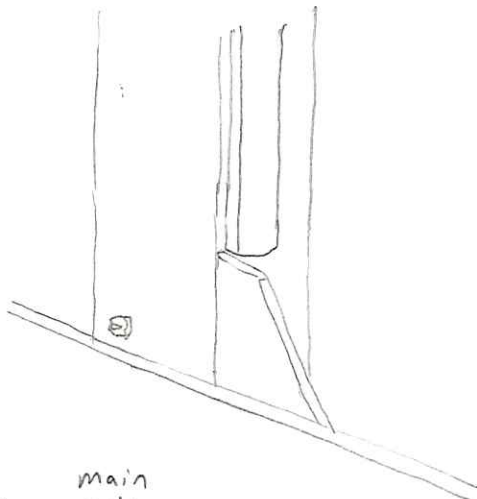
New Wheel Encoders

JHC 12/17/14

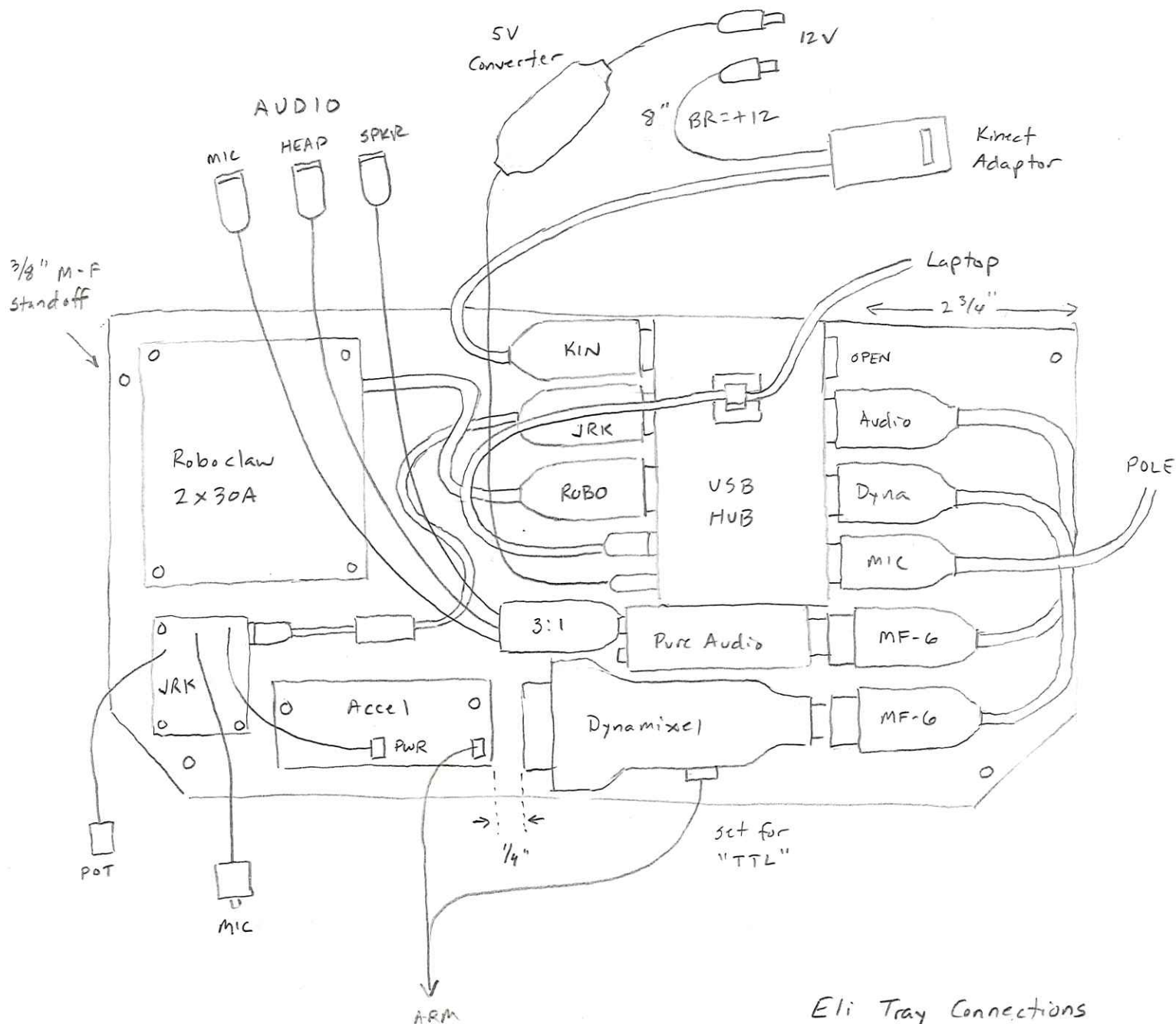




22 1/4" conduit



Eli's Bulk Cable Routing
JHC 2/16/12



Eli Tray Connections

JHC 10/18/12

